

Employment Protection and Tax Aggressiveness: Evidence from Wrongful Discharge Laws*

Douglas (DJ) Fairhurst, Yanguang Liu, and Xiaoran Ni

This version: July 2020

* Douglas (DJ) Fairhurst, dj.fairhurst@wsu.edu, Carson College of Business, Washington State University. Yanguang Liu, yliu46@email.arizona.edu, Eller College of Management, The University of Arizona. Xiaoran Ni, nxr@xmu.edu.cn, School of Economics & Wang Yanan Institute for Studies in Economics, Xiamen University. Liu is very grateful to Sandy Klasa for his numerous helpful suggestions and comments. We are grateful for constructive comments and discussions from Scott Cederburg, Shannon Chen, Yunsen Chen (discussant), Si Cheng (discussant), Katharine Drake, Kathleen Kahle, Christopher Lamoureux, Mollie Mathis (discussant), Richard Sias, Helen Wang (discussant), Ryan Williams, and participants in the CUFE-TiU PhD Program Workshop on Corporate Finance and FinTech (Beijing), the 2019 Frontiers of Business Research in China International Conference (Beijing), the 2019 Asian Finance Association Annual Meeting (Ho Chi Minh City), the 2019 American Accounting Association Annual Meeting (San Francisco), and the seminar at University of Arizona. All errors are our own.

Employment Protection and Tax Aggressiveness: Evidence from Wrongful Discharge Laws

Abstract:

We examine whether labor market frictions affect firms' tax aggressiveness. Exploiting the adoption of U.S. state-level Wrongful Discharge Laws as a quasi-exogenous shock to a firm's firing costs, we document a decline in tax aggressiveness for firms located in states that increase employment protection. We further show that greater employment protection increases distress risk. The decline in tax aggressiveness is more pronounced for firms that are more vulnerable to financial distress and constrained from external financial markets. Our results imply that firms avoid risky tax positions in order to mitigate increased distress risk due to more rigid labor costs.

Keywords: tax aggressiveness; employment protection; Wrongful Discharge Laws; firing costs; distress risk.

JEL: G31; G38; H26.

1. Introduction

Wages represent one of the largest portions of the cost of production for the average U.S. firm (Bai, Serfling, and Shaikh (2020)). Importantly, the nature of these labor costs has shifted dramatically because of regulation increasing employment protection. This shift resulted in more rigid labor costs and greater potential employee litigation, which has impacted firm-level financing, accounting, and investment decisions.¹ Yet, there has been no consideration of the impact of this shift in the nature of labor costs on another important cost faced by firms: the firm's tax expense. We look to fill this gap by considering how changes in labor costs impact firms' tax aggressiveness², given the importance of and uncertainty inherent in aggressive tax positions. For example, Blouin (2014) states that "aggressive tax planning stems from firms' appetite for risk."

At-will employment traditionally governed firing decisions in the U.S. Under at-will employment, employees can be terminated at any time for any reason. Over time, federal and state governments have adopted legal exceptions to at-will employment to protect employees from dismissal. While increased employment protection reduces the risk faced by employees, it makes the termination of employees more difficult. These rigid labor costs increase a firm's fixed costs and, as a result, operating leverage (Serfling (2016)). Increased operating leverage is associated with an increase in firm-level distress risk (e.g., Mandelker and Rhee (1984); Mauer and Triantis (1994); Kahl, Lunn, and Nilsson (2014)). Tax aggressiveness also involves risk. Under the theoretical model of Allingham and Sandmo (1972), tax positions are considered an investment with a risky payoff as there is

¹ Simintzi et al. (2015), Serfling (2016), Qiu (2019), Dang et al. (2019), Karpuz et al. (2019), Gao and Huang (2019), and Bai et al. (2020) show that employment protection affects capital structure, investments, risk management, payout policies, accounting conservatism, and real investment rates.

² Dyreng, Hanlon, and Maydew (2019) indicate that tax avoidance includes "a broad range of tax-reduction activities, ranging from benign tax-advantaged investments (e.g., tax-exempt municipal bonds) to aggressive strategies that might not be upheld if challenged". In this paper, we focus on tax aggressiveness, because the uncertainty inherent in these positions (Blouin (2014)) are more likely to be affected by a shift the nature of a firm's labor costs.

uncertainty regarding the likelihood the position would be challenged by tax authorities. Theoretically, it is unclear how greater distress risk stemming from more rigid labor costs affects tax aggressiveness decisions.

On one hand, greater employment protection could lead to more tax aggressiveness as firms may take more tax risk in response to greater distress risk. Classical risk-shifting theory and evidence suggest firms may substitute safer projects for riskier projects, especially as distress risk increases (Jensen and Meckling (1976); Eisdorfer (2008); Aretz et al. (2019)). Importantly, there is some empirical evidence of risk shifting in tax positions (e.g., Richardson, Lanis, and Taylor (2015)). The increased risk of distress from greater rigidity in labor costs may result in more aggressive tax positions.

On the other hand, greater employment protection could reduce tax aggressiveness. The theoretical framework of Almeida et al. (2011) makes opposite predictions of Jensen and Meckling (1976) of how firms' risk appetite changes as firms have greater distress risk. Specifically, in the model of Almeida et al. (2011) an increase in leverage leads firms to take safer, not riskier investments, especially for firms with constrained access to capital, as safer projects increase the probability of funding future investment. Increased firm-level operating leverage stemming from greater employment protection may disincentivize firms from taking aggressive tax positions, which potentially also increase risk (Ling et al. (2017); Heitzman and Ogneva (2019)). The resulting prediction is that an increase in employment protection results in lower tax aggressiveness because of increased operating leverage.

Ultimately, the effect of employment protection on a firm's tax aggressiveness is an empirical question. To address this question, we exploit the staggered adoption of Wrongful Discharge Laws (WDLs) as a quasi-natural shock to labor protection. WDLs move away from at-will employment by providing legal protection from termination in certain situations (Autor et al. (2006); Acharya et al. (2014); Serfling (2016)). WDLs matured into

three common laws that protect workers against different aspects of unjust dismissal: the Good Faith, Public Policy, and Implied Contract Exceptions. We focus on the adoption of the Good Faith Exception. This exception is applied in cases when a court determines that an employer discharged a worker out of bad faith, malice, or retaliation. Our analysis centers on this exception because it has a broad application in its deviation from at-will employment relative to the other two exceptions (e.g. Dertouzos and Karoly (1992); Kugler and Saint-Paul (2004)). However, we control for the other exceptions in all models. These laws apply to the location of the employee. We follow the literature and use headquarters locations employees are typically located in the headquarters state.

WDLs have several features that make them a viable quasi-natural experiment. These laws were adopted across several states over various time periods, which allow us to isolate the effects of the laws from macro-economic conditions. Also, WDLs are common law exceptions. As the decisions are made by judges in courts, they are less likely to be influenced by lobbying than laws created through legislation (Autor (2003)). Finally, several observations imply that WDLs impact firms substantially. Prior literature finds that employment levels, employment volatility, and firm entry decrease following its adoption (Dertouzos and Karoly (1992); Autor et al. (2007)). Firms also experience negative cumulative abnormal stock returns when their state adopts the law (Serfling (2016)). Further, an insurance market has developed to protect firms from the impacts of WDLs (Bai et al. (2020)).³

Our goal is to capture tax aggressiveness, or tax positions that are at risk of being reversed if a firm is questioned by tax authorities. We use discretionary book-tax

³ We note that the growth of the insurance market to protect firms largely occurred after our sample period and is unlikely to affect our results.

differences (*DDbTD*) as the primary measures of tax aggressiveness.⁴ Differences in financial accounting and tax income may be evidence of managing book earnings up or tax earnings down. *DDbTD* isolates the component of estimated book-tax differences that is not explained by total accruals, as the remaining difference likely stems from tax planning. If tax authorities use book-tax differences to identify aggressive tax positions, then this measure may correlate with the risk of a position being questioned (Mills (1998); Hanlon and Heitzman (2010)).

Our tests utilize a difference-in-differences approach in which the treatment and control groups consist of firms headquartered in states that have and have not adopted the Good Faith Exception, respectively. We find a decrease in tax aggressiveness, as measured by *DDbTD*, following the adoption of the Good Faith Exception. This finding is robust to the inclusion of firm, year, and industry-by-year fixed effects. Additionally, the relation is significant regardless of the presence of firm-level controls, suggesting our inference is not a result of including endogenous controls. Economically, the decrease in *DDbTD* corresponds to a 9.8% decline relative to the sample standard deviation, suggesting the effect is both statically and economically meaningful.⁵

A causal interpretation of the relation between the adoption of the Good Faith Exception and tax aggressiveness must satisfy the parallel trends assumption that the average change in aggressive tax positions would have been the same for both treatment and control firms absent the treatment. The evidence is consistent with this assumption. First, we find that shifts in tax aggressiveness occur one year after and not before the adoption of the exception, which alleviates concerns of reverse causality. Another concern is

⁴ *DDbTD* is a common measure of tax aggressiveness. See Desai and Dharmapala (2006), Frank et al. (2009), Chen et al. (2010), Rego and Wilson (2012), and Balakrishnan et al. (2018)).

⁵ The economic impacts are comparable to other studies. For example, Chen and Lin (2017) find that a reduction in analyst coverage is associated with an increase in *DDbTD* corresponding to 10.54% of the sample standard deviation for an average firm. Rego and Wilson (2012) find that the effect of a one standard deviation increase in equity risk incentives is to increase tax risk by 16.6%.

that the treatment and control firms differ in dimensions that impact tax aggressiveness. We address this issue by showing that the results are robust to using a matched sample. We match treatment firms to control firms based on firm-level characteristics. We use propensity score matching and continue to find a significant decrease in aggressive tax positions for treatment firms following the adoption of the Good Faith Exception. Last, we note that we are able to include state-by-year fixed effects when running our cross-sectional analyses and find consistent results. These fixed effects control for any state-level, time-varying factors that may be correlated with both the adoption of the law such as local GDP growth or the political environment at the time the laws are adopted. Collectively, these findings suggest that our results are robust to addressing concerns with the parallel trends assumption.

No single measure of tax aggressiveness captures all dimensions of aggressive tax positions (Hanlon and Heitzman (2010)). We document that the decrease in tax aggressiveness after the adoption is robust to a number of measures. The measures include total book-tax differences (without adjusted for accruals), a variable that estimates the probability that a firm uses a tax shelter, the unexplained portion of a firm's effective tax rate, and a measure of the difference between a firm's effective tax rate and the statutory tax rate.⁶ A robust relation across these various measures suggests that we are capturing various dimensions of tax aggressiveness.

Last, we address three alternative explanations for our findings. First, we confirm that our results are not explained by increased union power constraining tax aggressiveness (the main argument of Chyz et al. (2013)).⁷ Second, we confirm that the

⁶ We provide detailed explanation of variable construction in Appendix A. The results related to the use of alternative measures of tax aggressiveness are reported in Section 5.2.

⁷ As WDLs do not generally pertain to unionized employees, this channel is unlikely to hold (Autor (2003)). Nevertheless, we conduct additional tests to pin down this channel in Section 5.1.

main effect is not mechanical via decreased capital investment as a result of strengthened employee protection (Bai et al. (2020)). Last, we rule out patenting activity as an alternative explanation.

The finding that firms decreased tax aggressiveness following the adoption of the Good Faith Exception is consistent with the notion that greater employment protection increases firms' operating leverage through more rigid labor costs, increasing distress risk, and leading firms to take fewer aggressive tax positions, consistent with Almeida et al. (2011). We perform several tests to strengthen the tie to this theoretical channel and its underlying mechanisms. We first examine the direct link between the Good Faith Exception and distress risk. Using the adjusted Altman's Z-score and the expected default frequency (EDF) measure of Bharath and Shumway (2008), we show that greater employment protection increases distress risk and that this increase occurs only after the adoption of the law.

We next exploit various sources of cross-sectional variation in firm characteristics. We adopt the cohort-matching approach proposed by Gormley and Matsa (2011). Specifically, we treat each law passage as a cohort and pool the data of treated and control firms across cohorts to estimate the average treatment effect. This approach allows us to capture cross-sectional variation in firm characteristics in the year prior to the adoption.

The theoretical argument consistent with the paper's results is that firms reduce tax aggressiveness because of increased distress risk. We test whether firms with a higher probability of distress are more likely to reduce tax aggressiveness. We use several proxies for financial distress, including Altman's Z-Score and expected default frequency. We also use the proportion of short-term debt as greater shorter-term debt increases the exposure of firms to liquidity shocks (Custódio et al. (2013)). As our final measure of distress, we use industry cash flow volatility as firms in more volatile industries are more likely to need to

adjust employment in response to cash flow fluctuations (Klasa et al. (2018); Bai et al. (2020)), exposing these firms to a greater risk of distress. We find the relation between the Good Faith Exception and tax aggressiveness is most pronounced for firms with a greater distress risk across all four measures.

The theoretical rationale in Almeida et al. (2011) for firms reducing risk-taking in the presence of increased leverage is most pronounced for firms that face difficulty accessing external markets for future investment. This implies that the relation should be stronger for financially constrained firms. We find evidence consistent with this implication using three measures of financial constraints: whether the firm pays a dividend, the Whited-Wu index (Whited and Wu (2006)), and the Size-Age index (Hadlock and Pierce (2010)). In sum, we find that the increased rigidity in labor costs causes firms to reduce tax aggressiveness, which appears to be a result of firms reducing risk-taking in the presence of greater operating leverage.

Our paper contributes to work showing that employment laws can have a substantial effect on various corporate policies, financial reporting issues, and valuation (e.g., Matsa (2010); Garmaise (2011); Griffith and Macartney (2014); Simintzi et al. (2015); Loderer et al. (2016); Serfling (2016); Dang et al. (2019); Gao and Huang (2019); Karpuz et al. (2019); Qiu (2019)). Our evidence complements these studies by documenting that employment protection laws shape tax planning in addition to financing, investment, and accounting policies. An implication from our study is that government intervention in labor policies may unintentionally reduce firms' aggressive tax positions.

Our work also contributes to the ongoing debate of whether increased distress risk leads to risk-shifting or risk-mitigation. Classical risk-shifting theory suggests that firms near distress have incentives to substitute safer for riskier projects (Jensen and Meckling (1976); Eisdorfer (2008)). Yet, a growing literature indicates that the risk-mitigating

incentives can outweigh the risk-shifting incentives for firms near distress (Diamond (1989); Rauh (2009); Almeida et al. (2011); Gormley and Matsa (2011, 2016); Gilje (2016); Aretz et al. (2019)). We find that increased distress risk from greater labor protection leads to fewer risky tax positions, which provides support for the above risk-mitigating view.

2. Institutional background and hypothesis development

2.1 Institutional background

Wrongful discharge laws are exceptions to at-will employment that protect employees from unfair dismissal practices. The WDLs matured into three common law exceptions, the Good Faith, the Implied Contract, and the Public Policy Exception, which each differ in their application. The Good Faith Exception requires employers to treat workers in a fair manner. It applies in cases when an employer discharged a worker out of bad faith, malice, or retaliation, or, in other words, when the employee's termination was for something other than just cause. The Implied Contract Exception shields workers from termination when an employer has implied contractual provisions restricting employers' ability to fire employees. The Public Policy Exception protects employees from termination for refusing to violate a public policy (e.g., serve on a jury) or commit an illegal act.

Prior work (e.g. Dertouzos and Karoly (1992); Kugler and Saint-Paul (2004)) indicates that the Good Faith Exception represents the greatest deviation from at-will employment because it requires that termination must always be for just cause, leaving an ambiguous interpretation for courts. This exception is violated in 'timing' cases in which the employer intentionally fires employees before pension vests or employees receive their sales commissions. According to this exception, employees can seek compensation for both contractual losses and emotional and punitive damages. Emotional and punitive damages

can increase an employer's liability significantly and, there is no standard formula for a jury to determine the damages. These factors lead to uncertain settlement amounts.

By comparison, firms can mitigate the threat of lawsuits based on the Implied Contract Exception by designing personnel manuals and employees' handbooks that explicitly state that employment contracts are at-will (Miles (2000); Autor, Kerr, and Kugler (2007)). In other words, the effect of the law can be offset by careful contracting. As such, the adoption of Implied Contract Exception may not provide a drastic increase in employment protection.

Autor, Kerr, and Kugler (2007) also point out that the Public Policy Exception generally does not restrict employer behavior. Because courts limit public policy cases to clear violations of explicit public policy, rather than violations of an ambiguous sense of public obligation, the Public Policy Exception does not impose considerable constraints on employee termination. Therefore, we focus on the Good Faith Exception and treat the adoption of the Implied Contract and Public Policy Exceptions as additional control variables to examine the effect of employment protection on tax aggressiveness. Table 1 summarizes the dates when exceptions were passed by states.

[Insert Table 1 here]

An important assumption for the study is that, following the adoption of WDLs employment protection and expected firing costs increase due to an increase in expected litigation. We argue that it is the case in our setting for three reasons. First, outcomes in wrongful discharge lawsuits impose significant monetary costs on firms. Some studies document evidence of the significant costs and the average settlements are significant.⁸ For example, Dertouzos et al. (1988) analyze jury trials of wrongful discharge claims in

⁸ Westin et al. (1988) estimate that there were about 20,000 wrongful termination cases pending in state courts. The number of wrongful termination cases is likely higher since more states adopted WDLs.

California between 1980 and 1986. They find that plaintiffs won the majority of cases with the typical jury award of over \$650,000. Jung (1997) finds a lower percentage of successful lawsuits (46.5%) but a significantly higher average award (\$1.29 million). These estimates likely represent lower bound estimates as many cases are settled financially before ever reaching a jury.

Second, firms cannot completely offset the extent of losses related to employee lawsuits by purchasing employment practices liability insurance (EPLI) (Bai et al. (2020)). Although EPLI can provide insurance for losses from wrongful termination claims, the impact of EPLI on our findings is only limited because EPLI was not widespread in our sample period of 1970-2000. Additionally, the early application of EPLI often did not protect firms because no coverage was provided for “intentional acts.”

These impacts lead to the last observation of the impact on firms. Both firm policies and stock returns suggest that firms have largely been impacted by the increase in employment protection. Prior literature documents that employment levels, employment volatility, and firm entry decrease following its adoption (Dertouzos and Karoly (1992); Autor et al. (2007)). Firms also experience negative cumulative abnormal stock returns when their state adopts the law (Serfling (2016)).

We also note that, while it is likely that endogeneity concerns cannot be completely ruled out, it is unlikely that these concerns significantly cloud our inferences. The enactment of the Good Faith Exception is based on judicial rather than legislative decisions, which are more likely driven by the characteristics of the specific case and are less likely affected by political economy considerations, such as lobbying activities and omitted economic characteristics (Acharya et al. (2014)). In addition, Bai et al. (2020) show that economic factors were not major determinants of courts’ decisions to adopt the Good Faith Exception and therefore provide initial evidence that the recognition of the law likely

represents an exogenous shock. Further, changes in courts' position regarding the WDLs were unlikely to be anticipated by corporations, consistent with the abnormal market returns to court decisions (Serfling (2016)). Therefore, it's unlikely that firms altered policies in anticipation of the outcomes.

2.2 Hypothesis development

At-will employment has traditionally been the governing doctrine for firing decisions in the U.S. Under at-will employment, employees can be terminated with or without cause and at any time. Clearly, this policy benefits firms, providing great flexibility in labor decisions. Of course, this flexibility potentially comes at the cost of uncertainty and unfairness to employees.

Over time, federal and state governments have sought to balance these opposing forces by adopting legal exceptions to at-will employment that protect employees from dismissal in various situations. As a result, this increase in employment protection makes it more costly and difficult to terminate employees, making labor costs more fixed in nature. Rigid labor costs increase a firm's overall fixed costs. As a result, an increase in employment protection results in higher operating leverage (Serfling (2016)). Similar to financial leverage, increased operating leverage is associated with an increase in distress risk for firms (e.g., Mandelker and Rhee (1984); Mauer and Triantis (1994); Kahl, Lunn, and Nilsson (2014)). However, the effect of greater distress risk on tax aggressiveness is ambiguous.

Increased employment protection could lead to more tax aggressiveness if firms have an increased appetite for aggressive tax positions in the presence of greater distress risk. Tax positions can be considered an investment with a risky payoff as there is uncertainty regarding the likelihood the position would be challenged by tax authorities (Allingham and

Sandmo (1972)). Traditional theories of the risk-shifting theory suggest firms may substitute safer projects for riskier projects, especially as distress risk increases (e.g., Jensen and Meckling (1976)). Several empirical papers document evidence of risk shifting in investment in real assets (Eisdorfer (2008); Aretz et al. (2019)). There is also empirical evidence of risk shifting in tax positions (e.g., Richardson, Lanis, and Taylor (2015)). The implications of these models are that employment protection may lead firms to shift to riskier tax positions. This leads to our first hypothesis, stated in the alternative form:

H1. Increased employment protection results in higher tax aggressiveness in response to greater distress risk as firms potentially benefit from riskier tax positions while lenders bear the cost.

Alternatively, greater employment protection could lead to fewer risky tax positions. The predictions in the theoretical framework of Almeida et al. (2011) regarding firm-level risk appetites are opposite of the predictions in Jensen and Meckling (1976). In the model found in Almeida et al. (2011), firms approaching distress consider not only their current financial position but also their ability to meet future investment opportunities. As a result, firms approaching distress shift to shorter-term and safer investment projects to increase the ability to finance future investment projects.

An implication of their model is that an increase in distress risk coming through greater leverage leads to safer investments. For example, Gilje (2016) shows that oil and gas firms lower their risk in response to exogenous increases in distress risk. Relatedly, Aretz et al. (2019) indicate that increased distress risk due to hurricane strikes induces moderately (highly) distressed firms to increase (reduce) the risks of their operating segment portfolio.

This intuition extends to firms with a higher proportion of fixed operating costs. Increased operating leverage stemming from employment protection may disincentivize

firms from taking aggressive tax positions, which potentially also increase risk (Ling et al. (2017); Heitzman and Ogneva (2019)). The potential for safer projects in the presence of greater distress risk leads to our second hypothesis, also stated in the alternative form.

H2. Increased employment protection results in lower tax aggressiveness as firms reduce risky tax positions in response to greater distress risk to avoid distress costs and benefit from future financing opportunities.

3. Empirical design

3.1 Measuring tax aggressiveness

Tax avoidance includes “a broad range of tax-reduction activities, ranging from benign tax-advantaged investments (e.g., tax-exempt municipal bonds) to aggressive strategies that might not be upheld if challenged” (Dyreng, Hanlon, and Maydew (2019)). Increased distress risk from greater employment likely has little impact on tax positions that are clearly within the bounds of IRS rules. As such, we focus on tax aggressiveness, because of the uncertainty inherent in these positions (Blouin (2014)). These risky positions are theoretically more likely to be impacted by changing distress risk.

We focus on discretionary book-tax (*DBTD*) differences as our primary measure of tax aggressiveness, though we consider the robustness of our results to various other tax aggressiveness measures in Section 5.3. Firms report financial income under the requirement of Generally Accepted Accounting Principles (GAAP). Also, they must file income tax returns following the guidelines developed by the Internal Revenue Service (IRS). Because GAAP and tax regulations differ in a number of ways, the reported amounts of financial income and taxable income will differ. Subtracting taxable income from financial income, book-tax differences (BTD) are commonly used to capture the degree and riskiness of tax avoidance strategies. For example, Mills (1998) finds that firms with large

book-tax differences are more likely to be audited by the IRS and have larger proposed audit adjustments. Relatedly, Wilson (2009) finds that book-tax differences are larger for firms accused of engaging in tax shelters than for a matched sample of non-accused firms. Therefore, we build our measures on tax aggressiveness on various forms of book-tax differences.

However, book-tax differences may stem from tax avoidance or intentionally reducing taxable income. They may also stem from earnings management or opportunistically increasing financial income (Lim et al. (2011)). In the U.S., there is a nonconformity between financial reporting and tax reporting. Nonconformity means that there is no requirement that accounting and tax income correspond to one another in a given time period, suggesting that income can be managed up *while* taxes are being managed down (Frank et al. (2009)). To isolate the component of the *BTD* that is attributable to tax management, we construct *DDBTD* (Desai and Dharmapala (2006)). This measure is estimated through the following fixed effects regression:

$$BTD_{i,t} = \beta_1 TACC_{i,t} + \mu_i + \varepsilon_{i,t}, \quad (1)$$

where *BTD* is the total book-tax difference and *TACC* is the total accruals measured using the modified Jones (1991) Model each for firm *i* in period *t*. μ_i captures firm fixed effects that remove the average residual value across the sample period for each firm. By construction, this measure intends to isolate the component of the estimated book-tax difference that is not explained by accruals. One advantage of using *DDBTD* is that it reflects tax avoidance that generates both temporary and permanent book-tax differences

while some measures of tax aggressiveness do not.⁹ Given the seemingly permanent change in a firm's labor cost structure resulting from employment protection laws, firms would likely react with both temporary and permanent changes in tax aggressiveness.

3.2 Data and sample selection

Our primary sample utilizes financial statements and stock price data from Compustat. We remove firms in financial industries (SIC codes 6000–6999) and retain firms that have publicly traded stock, are incorporated in the United States, and have non-missing data for the main variables of interest. Following Autor et al. (2007), we begin our sample period in 1970. We end the sample period in 2000 to allow for two years following the final adoption of the Good Faith Exception by Louisiana in 1998. The main sample includes 57,467 firm-years. For our cross-sectional analyses, we also calculate the implied default probability, which requires the use of stock price data from the CRSP database. Wrongful discharge laws apply to the location of the employee. We follow the literature and use headquarters locations because employees are typically located in the headquarters state. We use headquarters location data from Compustat.

3.3 Empirical methodology

We adopt a difference-in-differences research design to examine the relation between the passage of the Good Faith Exception and corporate tax aggressiveness at the firm-year level. We estimate the following panel regression model:

$$TaxAgg_{ijst} = \alpha_1 Good\ Faith_{st} + X_{ist}\beta + v_i + \tau_t + \mu_j \times \tau_t + \varepsilon_{ijst}, \quad (2)$$

⁹ Temporary differences result from the differences in revenue and expense recognition policies when reporting the financial and taxable income. For example, bad debt expense can be recognized using the allowance method for financial reporting purposes while using the direct write-off method for tax purposes. By comparison, permanent differences result from items that (1) enter into pretax financial income but never into taxable income or (2) enter into taxable income but never into pretax financial income. For example, expenses incurred in obtaining tax-exempt income are recognized for financial reporting purposes but not for tax purposes.

where $TaxAgg_{ijst}$ is a firm's tax aggressiveness (primarily measured using *DDBTD*) for firm i in industry j , state s , and year t . $GoodFaith_{st}$ is an indicator variable that takes the value of one if a firm headquartered in state has adopted the Good Faith Exception as of year t , and zero otherwise. The coefficient on good faith gauges the effect of changes in state courts' position on the good faith on a firm's tax aggressiveness relative to that of firms in unaffected states. Under hypothesis 1 (2), the coefficient on this variable would be positive (negative). The regression model also includes a set of control variables X_{ijst} , firm fixed effects v_i , and year fixed effects w_t . The control variables also include $ImpliedContract_{st}$, and $PublicPolicy_{st}$ to control for the effect of the other two wrongful discharge laws. To alleviate the concern that industry-wide shocks can trigger both the adoption of labor laws and firms' tax aggressiveness decisions, we also control for industry (two-digit SIC)-times-year fixed effects in the model.¹⁰

The control variables include variables commonly found in tax aggressiveness regressions (e.g., Chyz et al. (2013); Chen and Lin (2017)).¹¹ Appendix A carefully defines each variable. We correct estimated standard errors in all regressions for heteroscedasticity and cluster at the state level. Given that the variation in the Good Faith Exception is at the state level, this clustering method accounts for potential time-varying correlations in unobserved factors that affect different firms within a given state (Bertrand et al. (2004)). This methodology also corrects for within-firm error term correlations over time and is therefore more general than firm-level clustering. Continuous variables, except economic variables, are winsorized at their 1st and 99th percentiles.

¹⁰ For example, a negative industry shock might affect the oil industry and result in massive layoffs that cause states to adopt wrongful discharge laws. Meanwhile, the negative industry shock might simultaneously lead firms in the oil industry to decrease tax aggressiveness or restrict the opportunities to reduce taxes. If so, the correlation observed in the present study cannot be interpreted as casual.

¹¹ We add variables common to both papers as our control variables. We note that Chyz et al. (2013) control for lagged values of the dependent variable while Chen and Lin (2017) do not. As robustness, we document that our results are also robust to the inclusion of lagged *DDBTD* in Table A1 of the internet appendix.

Table 2 presents summary statistics. In Panel A, we show that variables display distributional characteristics in line with past studies. In Panel B, we compare these variables for treated and control firms. Treatment and control firms differ significantly in a number of dimensions. We consider the differences between these two groups of firms by performing a matched sample analysis in Section 4.4.

[Insert Table 2 here]

4. Empirical results

4.1 Determinants of the recognition of Good Faith Exception

Before examining the relation between the recognition of Good Faith Exception and corporate tax aggressiveness, we address the concern that ex-ante tax aggressive activities of local firms may trigger the recognition of the Good Faith Exception in certain states. To test for these predictions, we follow Acharya et al. (2014) and Serfling (2016) and estimate a Cox proportional hazard model, where a failure event represents the adoption of the Good Faith Exception. States are excluded from the sample after they recognize the Good Faith Exception. All explanatory variables are lagged one year. Table 3 reports the estimation results.

In Column (1), we include the average level of tax aggressiveness in a state as the explanatory variable. The coefficient estimate is positive but insignificant, suggesting that state-level tax aggressiveness does not predict the recognition of the Good Faith Exception. In Column (2), we add a series of state-level measures of economic and political conditions, including dummies of the other two wrongful discharge laws (the Implied Contract and Public Policy exceptions), real GDP growth, the natural log of GDP, the unemployment rate, the natural log of the population, and union membership rate. In Column (3), we further

control for political balance.¹² The coefficient estimates of state-level tax aggressiveness are still insignificant, which serve as initial evidence that the recognition of Good Faith exceptions likely represent quasi-exogenous shocks with respect to corporate tax aggressiveness.

[Insert Table 3 here]

4.2 Main results

Table 4 reports the difference-in-difference estimates of the impact of the recognition of WDLs by state courts on the tax aggressiveness of firms in affected states. In Column (1), we examine the primary effect of Good Faith Exceptions by including firm- and year-fixed effects in the model without any control variables to ensure that potentially endogenous control variables do not bias the main effect. We find a negative and significant coefficient on *Good Faith*, suggesting that the adoption of the Good Faith exception leads to a reduction in tax aggressiveness.

In Column (2), we control for the effect of the vector of firm-level control variables and *Implied Contract* and *Public Policy* to ensure that omitted correlated variables do not drive the result. In Column (3), we further control for industry-times-year-fixed effects. In each case, we continue to find a negative and statistically significant impact of the Good Faith exception on tax aggressiveness. The *Good Faith* coefficient in Column (3) implies that the decrease in *DDBTD* relative to firms in non-adopting states is 9.8% of the sample standard deviation ($= -0.017/0.174$). In other words, the effect of Good Faith exception is

¹² Following Acharya et al. (2014), we obtain real GDP growth, the natural log of GDP, and the natural log of the population from the U.S. Bureau of Economic Analysis, unemployment rate from the U.S. Bureau of Labor Statistics, union membership rate from www.unionstats.com, and political balance from the Annual Statistical Abstracts of the U.S. Census Bureau. Political balance is the ratio of Democrat-to-Republican representatives in the Lower House (House of Representatives) for a given state and year; this variable is not available for the state of Nebraska, as it has a nonpartisan legislature (unicameral body) whose members are elected without party designation.

both economically and statistically meaningful.¹³

[Insert Table 4 here]

In sum, the preliminary evidence suggests that the increased distress risk stemming from greater employment protection leads to a decrease in tax aggressiveness. We next implement tests to address potential limitations on causal claims. We then consider evidence of the channels underlying the results.

4.3 Timing of changes in tax aggressiveness surrounding adoptions of WDLs

In Table 5, we consider the timing of the changes in tax aggressiveness relative to the adoption of the Good Faith exception. If reverse causality drives the results, we should observe a decreasing trend in the tax aggressiveness of firms in affected states prior to the adoption of the WDLs. This evidence would imply a violation of the parallel trends assumption that the trends in the tax aggressiveness of treatment firms in adopting states and control firms in non-adopting states are parallel prior to the adoption of WDLs.

For this test, we follow Serfling (2016) and decompose the key variable of interest into several variables to capture the timing of the adoption. The variables are *Good Faith Adoption*⁻¹, *Good Faith Adoption*⁰, *Good Faith Adoption*⁺¹, and *Good Faith Adoption*²⁺, which are equal to one if the firm is headquartered in a state that will adopt the Good Faith Exception in one year, adopts the Good Faith Exception in the current year, adopted the Good Faith Exception one year ago, or adopted the Good Faith Exception two or more years ago, respectively, and zero otherwise. The control variables in the three columns in Table 5 correspond to the control variables in Table 4.

Across all columns, the coefficients *Good Faith Adoption*⁻¹ are statistically

¹³ As *DDBTD* is a residual measure that centers around zero and the sample mean is negative, we do not compare the coefficient on *Good Faith* with the sample mean directly when interpreting the economic significance.

insignificant. In contrast, the coefficients on *Good Faith Adoption*²⁺ are significant across all columns, the directions of which are consistent with our findings in Table 4. These results indicate that firms in adopting states decrease their tax aggressiveness relative to control firms only after the adoption of the Good Faith Exception, but not before. Hence, reverse causality does not appear to explain our key result that an increase in employment protection triggers the firm's reduction of tax aggressiveness.

[Insert Table 5 here]

4.4 Matched sample analysis

We next consider whether the different reactions between treated and control firms stem from a difference in firm-level characteristics by performing a matched sample analysis. The procedures of selecting a control sample of firms follow Serfling (2016). We start by retaining all observations for treatment and control firms in year $t-1$ relative to the adoption of the Good Faith Exception. We treat each passage as a cohort. We require that both sets of firms have at least one observation in the ten-year window centered on the adoption of the law. We then use a logistic regression to estimate the probability of being a treated firm. When estimating these propensity scores, we include the vector of control variables from model 3 in Table 4 as independent variables as well as controlling for the industry- and cohort-fixed effects.

We match each treated firm in year $t-1$ to a control firm in year $t-1$ (without replacement) with the closest propensity score. We require the difference between propensity scores of the treated and control firm to be less than 0.01. We exclude all observations that do not satisfy the common support condition. We end up with 198 unique pairs of matched firms. Based on the full sample, we retain all observations of the matched pairs in the ± 5 years around the recognition of the Good Faith Exception. This results in

3,687 firm-year observations.

We run several tests to diagnose the quality of the match. In Table A2 of the Appendix, we report summary statistics of the propensity score matching procedures from three aspects. In Column (1), Panel A, we report the estimation on the baseline logistic model used to calculate propensity scores. A number of explanatory variable are significant in estimating the propensity scores. In Column (2), we then re-estimate the model based on the matched sample. The coefficients on all the explanatory variables except for *Public Policy* are no longer statistically significant. Further, the pseudo R-square shrinks from 0.199 to 0.053, and the p-value from the χ^2 test of overall model fitness changes from 0.000 to 0.846, suggesting that there are largely no different observable characteristics between the selected treatment and control firms after the matching procedure.

In Panel B, we report the distribution of the propensity scores for the post-match treatment and control groups and their differences and show that the two groups' propensity scores line up closely. In Panel C, we show that the sample means of the control variables for the matched treated and control firms are not significantly different from one another. In total, these tests indicate a successful match between treatment and control firms.

In Table 6, we re-estimate the baseline model based on the propensity-score matched sample. In Column (1), we use the model specification with the most stringent fixed effects (Column (3)) from Table 4. Using this matched sample, we continue to show a significant decline in tax aggressiveness for treated firms relative to control firms after the recognition of the Good Faith Exception. In Column (2) we check for the validity of the differences-in-differences setting by conducting the timing test based on the matched sample. Specifically, we replace the indicator variable capturing the adoption of the Good Faith Exception with

the four timing variables used in Table 5. The results imply that there is no trend of declining tax aggressiveness before the recognition of the Good Faith Exceptions, and the decline in tax aggressiveness emerges only after the legal changes. Taken together, the timing test and the matched sample analysis support the causal interpretation of the impact of the adoption of the Good Faith Exception on tax aggressiveness.

[Insert Table 6 here]

4.5 Employment protection, tax aggressiveness, and distress risk

The evidence of a negative relation between increased employment protection and tax aggressiveness is consistent with the theoretical model of Almeida et al. (2011). Under this interpretation, greater employment protection increases distress risk through higher operating leverage and firms reduce aggressive tax positions in response to greater distress risk. In this section we run several tests to document this underlying mechanism.

4.5.1 The effect of employment protection on distress risk

A prerequisite condition for the underlying channel to hold is that the recognition of Good Faith Exceptions increases the distress risk of treated firms. While a number of papers have argued that heightened labor protection increases distress risk by increasing operating leverage, we explicitly examine the direct link between Good Faith Exceptions and distress risk in Table 7.

We employ two measures of distress risk. The first measure is *Zscore*, the adjusted Altman's Z-score constructed following Serfling (2016), a higher level of which indicates a lower likelihood of default. The second measure is the expected default frequency (EDF) measure of Bharath and Shumway (2008), which is a simplified version of Merton (1974) structural distance-to-default model (Brogaard et al. (2017)). A higher level of EDF indicates a higher likelihood of default.

We regress these two measures on the Good Faith exception and control variables in Columns (1) and (2). The coefficients on Good Faith negatively related to *Z-Score* and positively related to *Expected Default Frequency*, and each relation is significant. This evidence indicates that greater employment protection is associated with higher distress risk. Based on the sample mean of *Zscore* and *Expected Default Frequency* (1.483 and 0.093, respectively), the recognition of Good Faith Exception increases the distress risk of an average treated firm by 8.4% ($=0.124/1.483$) to 16.1% ($=0.015/0.093$) relative to the sample mean compared to control firms. In Columns (3)-(4), the results imply that there is no trend of increasing distress risk before the recognition of the Good Faith Exceptions and the increases in distress risk emerge only after the legal change. These findings corroborate the implication of the underlying mechanism that greater operating leverage stemming from employment protection results in greater distress risk. In the next section, we will provide further evidence on this issue through cross-sectional analyses.

[Insert Table 7 here]

4.5.2 Cross-sectional analyses

We consider further evidence of the underlying mechanism using two sets of cross-sectional analyses. Specifically, we interact *Good Faith* with various firm characteristics based on predicted cross-sectional variation implied by the theoretical models that our main results are consistent with. The primary purpose of these tests is to shed further light on the economic mechanism behind our results and show that greater employment protection reduces tax aggressiveness by increasing the risk of employee litigation and cost associated with tax planning. In addition, the tests also strengthen our causal claims in two ways. First, if a variable omitted from our main regression model were to drive the main results, this variable would also have to explain the cross-sectional results we report here. Second,

as described below, we use a cohort approach for cross-sectional analysis. This approach allows for the use of state-year fixed effects. If state-level omitted variables are uncorrelated with the interaction term, these results should be unaffected by omitted state-level factors (see Bai et al. (2020)).

We adopt the cohort-matching approach proposed by Gormley and Matsa (2011) for these cross-sectional analyses. Specifically, we treat each law passage as a cohort, and then pool the data of treated and control firms across cohorts to estimate the average treatment effect. Our main specifications rely on *Good Faith* as an indicator of increased employment protection. This variable is essentially an interaction between whether a firm has adopted the Good Faith Exception and whether it is after the year in which the firm adopted the law. If we were to interact this variable with firm-level characteristics, we would essentially be using a triple interaction without controlling for some cross-interaction terms (e.g., the ‘treated’ variable and the firm characteristic). The stacked approach allows for triple interactions with the cross terms being absorbed in the fixed effects. This approach also allows us to compare $t-1$ firm-level characteristics because control firms are selected as part of a cohort. The model is specified as follows:

$$TaxAgg_{ijcst} = \gamma_0 Good\ Faith_{st} + \gamma_1 Good\ Faith_{st} \times X_{ic} + \gamma_2 X_{ic} + \omega_c \times \tau_t + \omega_c \times v_i + \mu_j \times \tau_t + \sigma_s \times \tau_t + \varepsilon_{it} \quad (3)$$

where X_{ic} is one of several variables used to identify cross-sectional differences constructed for each firm in each cohort.

We control for cohort-times-year-fixed effects ($\omega_c \times \tau_t$), cohort-times-firm-fixed effects ($\omega_c \times v_i$), industry-times-year-fixed effects ($\mu_j \times \tau_t$), and state-of-headquarter-times-year-fixed effects ($\sigma_s \times \tau_t$) in the model. In this approach, we allow the firm- and year-fixed effects to vary by cohort, which is more conservative than including simple fixed effects.

Also, we deliberately do not control for any time-varying accounting variables in these regressions because these variables may be affected by the adoption of the Good Faith Exception, and including them may confound estimates of $Good\ Faith_{st} \times X_{ic}$, the variable of interest.¹⁴

We first focus on measures of distress risk. If the reduction in tax aggressiveness stems from increased distress risk, then the effect should be most pronounced for firms with high distress risk. We use several proxies for financial distress risk. First, we use the two distress risk measures used in Table 7: Altman's Z-Score and the *expected default frequency*. We also use the proportion of short-term debt because shorter-term debt increases the exposure of firms to liquidity shocks (Custódio et al. (2013)). Last, we use industry cash flow volatility as firms in more volatile industries are more likely to need to adjust employment in response to cash flow fluctuations (Klasa et al. (2018); Bai et al. (2020)), exposing these firms to greater risk of distress.

The results are tabulated in Table 8. Columns (1), (2), (3), and (4) capture cross-sectional variation with respect to modified Altman's Z-Score, expected default frequency, short-term debt, and cash flow volatility, respectively. We find the relation between the Good Faith Exception and tax aggressiveness is most pronounced for firms with greater distress risk across all four measures.¹⁵ This evidence provides greater evidence that distress risk underlies the reduction in tax aggressiveness.

[Insert Table 8 here]

The theoretical rationale in Almeida et al. (2011) for firms reducing risk taking in the presence of greater distress risk is most pronounced for firms that face difficulty

¹⁴ As robustness, we do add controls to these models and find that our results are robust to the inclusion of these controls. These results, which are tabulated in Tables A3 and A4 of the internet appendix, suggest that our control variables do not confound the cross-sectional results.

¹⁵ As we control for cohort-times-firm-fixed effects and state-of-headquarter-times-year-fixed effects, the level terms for Good Faith and the firm-level characteristics are absorbed and therefore are not reported in the table.

accessing external markets for future investment. This implies that the relation should be stronger for firms with costly access to these markets, or financially constrained firms. Here, we use three common measures of financial constraints: an indicator variable that captures firms that do not pay a dividend, the Whited-Wu (WW) index (Whited and Wu (2006)), and the Size-Age (SA) index (Hadlock and Pierce (2010)).

These tests are tabulated in Table 9. For each proxy of financial constraints, the negative relation between the adoption of the Good Faith Exception and tax aggressiveness is pronounced for financially constrained firms (i.e., firms that do not pay a dividend and firms with higher WW index or SA index values), again consistent with the model of Almeida et al. (2011). In sum, we find that the increased rigidity in labor costs causes firms to reduce tax aggressiveness, which appears to be a result of firms reducing risk-taking in the presence of greater operating leverage.

[Insert Table 9 here]

5. Additional analysis and robustness

5.1 Alternative explanations

Thus far, we have provided fruitful evidence that employment protection discourages tax aggressiveness. However, several alternative explanations may explain our results. In this section, we briefly discuss three alternative explanations, namely the strengthened union power, increased innovation, and the increased capital expenditures, and then provide the results of tests that suggest our findings are not a product of these alternative explanations. We report the estimation results from these tests in Table 10.

In Panel A, we address the issue that our main results can be explained by strengthened union power that constrains tax aggressiveness, as Chyz et al. (2013) find a negative association between firms' tax aggressiveness and union power. Greater

employment protection can have a significant effect on real outcomes such as wages, employment, and financing decisions (Serfling (2016)). As WDLs do not generally pertain to unionized employees (Autor (2003)), this channel is unlikely to hold. Nevertheless, we still consider whether the effect of labor union drives our results in the event that the adoption of the Good Faith Exception correlates with greater union power.

In Column (1), we introduce an indicator variable *Right-to-Work Law* to measure ex-ante union bargaining power. State-level variation in the presence of Right-to-Work Laws provides exogenous variation in the strength of labor unions (John et al. (2015)). In states with Right-to-Work Laws, employees are not required to enter into agreements which require them to join a union in order to get or keep a job. This weakens the position of labor unions in these states. Thus, labor unions are stronger in firms located in states *without* Right-to-Work Laws. As the adoption of Right-to-Work Laws constrains union bargaining power, we conjecture that if this channel holds, the decrease in tax aggressiveness should be less pronounced for firms located in states that have adopted Right-to-Work Laws. However, the results fail to exhibit significant cross-sectional differences in the effect of employment protection with respect to the adoption of such laws.

In Columns (2)-(3), to further examine whether union channel holds, we follow Klasa et al. (2009) and use two state-level union coverage rates to measure unionization, which represent the percentage of total workers who are covered by unions in collective bargaining agreements in each state.¹⁶ If the union channel holds, we expect that the decrease in tax aggressiveness should be more pronounced for firms located in states with high union coverage rate. However, the estimation results show that the coefficients on

¹⁶ This data is available at www.unionstats.com, which is maintained by Prof. Barry Hirsch and Prof. David Macpherson.

both interaction terms are not significant¹⁷. Across all three measures, the results fail to exhibit significant cross-sectional differences in the effect of employment protection with respect to labor union coverage levels, suggesting that this channel is unlikely to drive our main results.

In Panel B, we first address the concern that the impact of the Good Faith Exception on tax aggressiveness is mechanically driven by the relationship between the Good Faith Exception and capital expenditures. Recent evidence shows that capital expenditures decline following the adoption of the Good Faith Exception (Bai et al. (2020)). To the extent that the decrease in capital expenditures reduces depreciation, which may lead to an increase in pre-tax income, this may result in a mechanical increase in tax expense and a perceived decrease in tax aggressiveness.

We use two tests to address this potential mechanical relation. First, we conjecture that if our findings are mechanically driven by the decrease in capital expenditures, then the main effect will be more pronounced for firms where there are substantial *decreases* in investment around the adoption of the Good Faith Exception. To test the above conjecture, we examine the effect of the Good Faith Exception for firms that experience a decrease in capital expenditures from year $t-1$ to year t . We also test the effect of the Good Faith Exception for the firms that increase investment. In Columns (1)-(2), the results show that the baseline relation is statistically significant and negative for both subgroups of firms, suggesting that the change in tax aggressiveness is not a mechanical by-product of firms decreasing their total investment.

Second, we consider whether changes in tax aggressiveness are either positively or negatively related to capital expenditures. That is, if the decline in tax aggressiveness is

¹⁷ As the variation of Right-to-Work Laws and union coverage rates lies at the state-year level, to obtain an estimation on the interaction term of Good Faith and those measures, state-by-year fixed effects are not included in the model.

mechanically driven by the decrease in capital expenditures, the main effect should concentrate in firms where there is a *positive* correlation between the changes in tax aggressiveness and those in capital expenditures. To test this, we calculate the one year change in both CapEx and *DDBTD* from year $t-1$ to t and split our sample into two groups: those with a positive correlation between these two changes and those with a negative correlation.

In Columns (3)-(4), we run our main model for the two sub-samples of firms. We observe a negative and significant effect of the Good Faith Exception in both sub-samples, regardless of the correlation between the changes in tax aggressiveness and those in capital expenditures. Overall, these results indicate that our evidence of the impact of WDLs on tax aggressiveness is unlikely to be mechanically driven by the relationship between those laws and firm capital expenditures.

We also consider the possibility that our main findings might be driven by innovative firms that have significant patenting activities. Acharya et al. (2014) find that the adoption of Wrongful Discharge Laws leads to an increase in innovation, as these laws solve a labor hold-up problem. Higgins, Omer, and Phillips (2015) provide evidence that ‘prospecting’ firms, such as those that are more innovative, differ in the extent to which they manage taxes. As such, an alternative explanation for our results is that the increase in innovative activity also affects tax aggressiveness.

To consider this possible mechanism, we split the sample based on whether a firm has patents in year $t-1$. Patent data are from the NBER patent citation database assembled by Hall, Jaffe, and Trajtenberg (2001). We find that the main effect is more prominent in the subsample with no patent applications, suggesting that patenting activities are unlikely to drive our main findings.

[Insert Table 10 here]

5.2 Alternative measures of tax aggressiveness

In Table 11, we examine whether our main findings hold when using alternative measures of corporate tax aggressiveness. We employ four additional measures including *BTD*, *SHELTER*, *DTAX*, and *PERMDIFF*. *BTD* is the total book-tax difference. This measure has the same intuition as *DDBTD*. However, this measure does not difference out the portion of the total book-tax difference that is attributable to earnings management, and, as such, ensures that our results are robust to potential errors in this differencing process. Following Wilson (2009), we construct a tax-sheltering probability measure, *SHELTER*. This measure estimates activity in tax-sheltering activity, a specific yet important type of tax aggressiveness, by using firm-level characteristics and estimated coefficients from a tax shelter determinants model (see Appendix A for more details). Frank et al. (2009) argue that it is the permanent component of book-tax differences that is significantly related to actual cases of tax sheltering and likely engenders greater risks. As such, we also consider two measures of permanent book-tax differences. The first is *DTAX*, the discretionary permanent book-tax difference. This measure is the residual from permanent differences regressed on characteristics that affect the likelihood of permanent differences. We also employ *PERMDIFF* without differencing out the determinants as an additional measure of tax aggressiveness.

In Columns (1)-(4), the findings indicate that the recognition of Good Faith Exceptions is associated with declined tax aggressiveness across each of the measures of tax aggressiveness, which is consistent with the main findings. In Columns (5)-(8), we confirm that the decrease in tax aggressiveness emerges only after the recognition of Good Faith Exception, which further suggests a causal interpretation. In sum, the decline in tax

aggressiveness does not appear to be a product of the measure of tax aggressiveness used. Instead, this table suggests that we broadly capture tax aggressiveness across several proxies.

[Insert Table 11 here]

5.3 Additional Robustness Checks

Last, we check for the robustness of the relation between employment protection and corporate tax aggressiveness, and report estimation results in Table 12. For each test, we use all five measures of tax aggressiveness. In Panel A, we specify our main models using the cohort-matched sample in order to alleviate the omitted variable problem through controlling for a series of high-degree fixed effects. In Panel B, we control for macro-level omitted variables by controlling for region-times-year-fixed effects in the model.¹⁸ California and Massachusetts have all three WDLs in place from the late 1980s and onward, offering their employees significant protection against unjust dismissal (Acharya et al. (2014)). In Panel C, we exclude firms headquartered in these two states to make sure that our findings are not driven entirely by these two states. For each of these three tests, we continue to find significant results across all five measures of tax aggressiveness.

Throughout the paper we use headquarters locations that are based only on the most recent headquarters location, a feature of the Compustat database. This backfilled data may be a noisy proxy for actual locations. In Panel D, we exclude firms with growth exceeding 100%, which may have experienced events that initiated changes in their headquarters locations (Serfling (2016)). In Panel E, we exclude firms that have changed

¹⁸ We follow the traditional classification and classify firms' headquarter states into 6 regions: (1) New England (including CT, MA, ME, NH, RI, and Vermont); (2) Mid-Atlantic (including DC, DE, MD, NJ, NY, and PA); (3) Midwest (including IA, IL, IN, KS, MI, MN, MO, ND, NE, OH, SD, and WI); (4) West (including CA, CO, ID, MT, NV, OR, UT, WA, and WY); (5) South (including AL, AR, FL, GA, KY, LA, MS, NC, SC, TN, VA, and WV); (6) Southwest (including AZ, NM, OK, and TX).

their headquarters throughout our sample period. To identify headquarters location switches, we use the data provided by Bai et al. (2020).¹⁹ This data is based on EDGAR filings for years 1996 and later and hand-collected data prior to 1996. We find that, whether we drop firms with large sales growth or drop firms for which we observe headquarters relocations, we continue to find a significant negative relation between the adoption of the Good Faith Exception and tax aggressiveness.

[Insert Table 12 here]

6. Conclusion

This paper exploits exogenous increases in employment protection arising from the adoption of U.S. state-level employment protection laws (particularly, the Good Faith Exception) to examine how greater employment protection affects corporate tax aggressiveness. We find a significant decrease in tax aggressiveness following the adoption of these laws. This finding is consistent with the notion that employment protection increases distress risk through more rigid labor costs, and, as a result, firms' appetites for risky tax positions decrease.

Supporting the above notion, we show that the decline in tax aggressiveness after the recognition of the Good Faith Exception is more pronounced for firms that are likely to be affected by an increase in distress risk. Specifically, we observe a greater decline in tax aggressiveness for firms that already have high distress risk. Further, firms that face high costs in accessing external financial markets also are more likely to reduce tax aggressiveness.

Overall, our paper fills a gap in the literature by demonstrating how the changing nature of a significant operational cost, namely labor costs, impacts the tendency of firms to

¹⁹ This data is available at <https://academic.oup.com/rfs/article/33/2/644/5522377#supplementary-data>.

be aggressive in their tax positions. Our findings highlight the interplay between labor costs and firms' approaches to taking tax positions that may be questioned by tax authorities. Further, our results document that a likely unintended side effect of stronger employment protection is reduced aggressive tax avoidance.

References

- Acharya, V. V., Baghai, R. P., and Subramanian, K. V. (2014). Wrongful discharge laws and innovation. *The Review of Financial Studies*, 27(1), 301-346.
- Allingham, M. G., and Sandmo, A. (1972). Income tax evasion: A theoretical analysis. *Journal of Public Economies*, 1, 323-338.
- Almeida, H., Campello, M., and Weisbach, M. S. (2011). Corporate financial and investment policies when future financing is not frictionless. *Journal of Corporate Finance*, 17(3), 675-693.
- Aretz, K., Banerjee, S., and Pryshchepa, O. (2019). In the path of the storm: does distress risk cause industrial firms to risk-shift?. *Review of Finance*, 23(6), 1115-1154.
- Autor, D. H. (2003). Outsourcing at will: The contribution of unjust dismissal doctrine to the growth of employment outsourcing. *Journal of Labor Economics*, 21(1), 1-42.
- Autor, D. H., Donohue III, J. J., and Schwab, S. J. (2006). The costs of wrongful-discharge laws. *The Review of Economics and Statistics*, 88(2), 211-231.
- Autor, D. H., Kerr, W. R., and Kugler, A. D. (2007). Does employment protection reduce productivity? Evidence from US states. *The Economic Journal*, 117(521), F189-F217.
- Balakrishnan, K., Blouin, J., and Guay, W. (2019). Tax Aggressiveness and corporate transparency. *The Accounting Review*, 94(1), 45-69.
- Bai, J. J., Fairhurst, D. J., and Serfling, M. (2020). Employment protection, investment, and firm growth. *The Review of Financial Studies* 33(2), 644-688.
- Bai, J. J., Serfling, M., and Shaikh, S. (2020). Financial Disclosure Transparency and Employee Wages. University of Tennessee Working Paper.
- Bertrand, M., Duflo, E., and Mullainathan, S. (2004). How much should we trust differences-in-differences estimates?. *The Quarterly Journal of Economics*, 119(1), 249-275.
- Balakrishnan, K., Blouin, J., and Guay, W. (2019). Tax Aggressiveness and Corporate Transparency. *The Accounting Review*, 94(1), 45-69.
- Bharath, S. T., and Shumway, T. (2008). Forecasting default with the Merton Distance to Default model. *Review of Financial Studies*, 21(3), 1339-1369.
- Blouin J. (2014). Defining and measuring tax planning aggressiveness. *National Tax Journal*, 67(4): 875.
- Brogaard, J., Li, D., and Xia, Y. (2017). Stock liquidity and default risk. *Journal of Financial Economics*, 124(3), 486-502.
- Chen, T., and Lin, C. (2017). Does Information Asymmetry Affect Corporate Tax Aggressiveness?. *Journal of Financial and Quantitative Analysis*, 52(5), 2053-2081.
- Chyz, J. A., Leung, W. S. C., Li, O. Z., and Rui, O. M. (2013). Labor unions and tax aggressiveness. *Journal of Financial Economics*, 108(3), 675-698.

- Chen, S., Chen, X., Cheng, Q., and Shevlin, T. (2010). Are family firms more tax aggressive than non-family firms?. *Journal of Financial Economics*, 95(1), 41-61.
- Custodio, C., Ferreira, M. A., and Laureano, L. (2013). Why are U.S. firms using more short-term debt?, *Journal of Financial Economics*, 108(1), 182-212.
- Dang, V. A., De Cesari, A., and Phan, H. V. (2019). Employment protection and share repurchases: Evidence from Wrongful Discharge Laws. Available at SSRN 3188011.
- Desai, M. A., and Dharmapala, D. (2006). Corporate tax avoidance and high-powered incentives. *Journal of Financial Economics*, 79(1), 145-179.
- Dertouzos, J. N., Holland, E., and Ebener, P. A. (1988). The legal and economic consequences of wrongful termination. Rand Corporation.
- Dertouzos, J. N., and Karoly, L. A. (1992). Labor market responses to employer liability. Rand Corporation.
- Diamond, D. W. (1989). Reputation acquisition in debt markets. *Journal of Political Economy*, 97(4), 828-862.
- Dyreng, S. D., Hanlon, M., and Maydew, E. L. (2010). The effects of executives on corporate tax avoidance. *The Accounting Review*, 85(4), 1163-1189.
- Dyreng, S. D., Hanlon, M., and Maydew, E. L. (2019). When Does Tax Avoidance Result in Tax Uncertainty? *The Accounting Review*, 94(2), 179-203.
- Eisdorfer, A. (2008). Empirical evidence of risk shifting in financially distressed firms. *Journal of Finance*, 63(2), 609-637.
- Frank, M. M., Lynch, L. J., and Rego, S. O. (2009). Tax reporting aggressiveness and its relation to aggressive financial reporting. *The Accounting Review*, 84(2), 467-496.
- Gao, H., and Huang, J. (2019). Employee Firing Costs and Accounting Conservatism: Evidence From Wrongful Discharge Laws. Available at SSRN 3276474.
- Garmaise, M. J. (2011). Ties that truly bind: Noncompetition agreements, executive compensation, and firm investment. *The Journal of Law, Economics, and Organization*, 27(2), 376-425.
- Gilje, E. P. (2016). Do firms engage in risk-shifting? Empirical evidence. *Review of Financial Studies*, 29(11), 2925-2954.
- Gormley, T. A., and Matsa, D. A. (2011). Growing out of trouble? Corporate responses to liability risk. *Review of Financial Studies*, 24(8), 2781-2821.
- Gormley, T. A., and Matsa, D. A. (2016). Playing it safe? Managerial preferences, risk, and agency conflicts. *Journal of Financial Economics*, 122(3), 431-455.
- Griffith, R., and Macartney, G. (2014). Employment protection legislation, multinational
- Hadlock, C. J., and Pierce, J. R. (2010). New evidence on measuring financial constraints: Moving beyond the KZ index. *Review of Financial Studies*, 23(5), 1909-1940.
- Hall, B., Jaffe, A., and Trajtenberg, M.. 2001. The NBER patent citations data file: Lessons, insights and methodological tools. Working Paper, NBER.

- Hanlon, M., and Heitzman, S. (2010). A review of tax research. *Journal of Accounting and Economics*, 50(2), 127-178.
- Heitzman S M, Ogneva M. (2019). Industry tax planning and stock returns. *The Accounting Review*, 94(5), 219-246.
- Higgins, D., Omer, T. C., and Phillips, J. D. (2015). The influence of a firm's business strategy on its tax aggressiveness. *Contemporary Accounting Review* 32(2), 674-702.
- Jensen, M. C., and Meckling, W. H. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. *Journal of Financial Economics*, 3(4), 305-360.
- Jung, D.J. (1997). Jury verdicts in wrongful termination cases. Public Law Research Institute Report, University of California Hastings College of the Law.
- John, K., Knyazeva, A., and Knyazeva, D. (2015). Employee rights and acquisitions. *Journal of Financial Economics*, 118(1), 49-69.
- Jones, J. J. (1991). Earnings management during import relief investigations. *Journal of Accounting Research*, 29(2), 193-228.
- Kahl, M., Lunn, J., and Nilsson, M. (2014) Operating leverage and corporate financial policies, Working paper, University of Colorado.
- Karpuz, A., Kim, K., and Ozkan, N. (2020). Employment protection laws and corporate cash holdings. *Journal of Banking & Finance*, 111, 105705.
- Klasa, S., Ortiz-Molina, H., Serfling, M., and Srinivasan, S. (2018). Protection of trade secrets and capital structure decisions. *Journal of Financial Economics*, 128(2), 266-286.
- Klasa, S., Maxwell, W. F., and Ortiz-Molina, H. (2009). The strategic use of corporate cash holdings in collective bargaining with labor unions. *Journal of Financial Economics*, 92(3), 421-442.
- Kugler, A. D., and Saint-Paul, G. (2004). How do firing costs affect worker flows in a world with adverse selection? *Journal of Labor Economics*, 22(3), 553-584.
- Lim, Y. (2011). Tax avoidance, cost of debt and shareholder activism: Evidence from Korea. *Journal of Banking & Finance*, 35(2), 456-470.
- Ling, C., Maydew, E. L., Zhang, L., and Zuo, L. (2017). Customer–supplier relationships and corporate tax avoidance. *Journal of Financial Economics* 123(2), 377-394.
- Loderer, C., Waelchli, U., and Zeller, J. (2017). Employment protection and investment opportunities. European Corporate Governance Institute Working Paper.
- Mandelker, G. N., and Rhee, S. G. (1984). The Impact of the Degrees of Operating and Financial Leverage on Systematic Risk of Common Stock. *Journal of Financial and Quantitative Analysis*, 19(1), 45-57.
- Mauer, D. C., and Triantis, A. J. (1994). Interactions of Corporate Financing and Investment Decisions: A Dynamic Framework. *Journal of Finance*, 49(4), 1253-1277.
- Merton R. C. (1974). On the pricing of corporate debt: The risk structure of interest rates. *Journal of Finance*, 29(2), 449-470.

- Matsa, D. A. (2010). Capital structure as a strategic variable: Evidence from collective bargaining. *The Journal of Finance*, 65(3), 1197-1232.
- Mills, L. F. (1998). Book-tax differences and Internal Revenue Service adjustments. *Journal of Accounting research*, 36(2), 343-356.
- Miles, T. J. (2000). Common law exceptions to employment at will and US labor markets. *Journal of Law, Economics, and Organization*, 16(1), 74-101.
- Qiu, Y. (2019). Labor adjustment costs and risk management. *Journal of Financial and Quantitative Analysis*, 54(3), 1447-1468.
- Rauh, J. D. (2009). Risk shifting versus risk management: Investment policy in corporate pension plans. *Review of Financial Studies*, 22(7), 2687-2733.
- Rego, S. O., and Wilson, R. (2012). Equity risk incentives and corporate tax aggressiveness. *Journal of Accounting Research*, 50(3), 775-810.
- Richardson, G., Lanis, R., and Taylor, G. (2015). Financial distress, outside directors and corporate tax aggressiveness spanning the global financial crisis: An empirical analysis. *Journal of Banking & Finance*, 52, 112-129.
- Serfling, M. (2016). Firing costs and capital structure decisions. *The Journal of Finance*, 71(5), 2239-2286.
- Simintzi, E., Vig, V., and Volpin, P. (2014). Labor protection and leverage. *The Review of Financial Studies*, 28(2), 561-591.
- Westin, A. F., and Feliu, A. G. (1988). *Resolving employment disputes without litigation*. BNA Books.
- Whited, T. M., and Wu, G. (2006). Financial constraints risk. *Review of Financial Studies*, 19(2), 531-559.
- Wilson, R. J. (2009). An examination of corporate tax shelter participants. *The Accounting Review*, 84(3), 969-999.

Appendix A Variable Definitions

Variable	Definition
<i>Measures of Tax Aggressiveness</i>	
DDBDT	Desai and Dharmapala's (2006) residual book-tax difference, which equals the residual from the following fixed effects regression: $BTD_{i,t} = \beta_1 tacc_{i,t} + \mu_i + \varepsilon_{i,t}$, where BTD is the total book-tax difference, and $tacc$ is the total accruals measured using the cash flow method of Hribar and Collins (2002).
BTD	Total book-tax difference is book income less taxable income scaled by lagged assets (at). Book income is pretax income (pi) in year t . Taxable income is the sum of current federal tax expense ($txfed$) and current foreign tax expense ($txfo$) divided by the statutory tax rate minus the change in net operating loss carryforwards ($tlcf$) in year t . If the current federal tax expense is missing, the total current tax expense equals the total income taxes (txt) minus deferred taxes ($txdi$), state income taxes (txs), and other income taxes (txo).
SHELTER	Estimated probability that a firm engages in a tax shelter based on Wilson's (2009) tax-sheltering model: $SHELTER = -4.86 + 5.20 \times BTD + 4.08 \times dac - 1.41 \times Leverage + 0.76 \times at + 3.51 \times ROA + 1.72 \times fi + 2.43 \times R\&D$, where BTD is the total book-tax difference; da is the absolute value of discretionary accruals from the modified cross-sectional Jones (1991) model; at is the total assets; ROA is the return on assets, measured as operating income (pi - xi) divided by lagged assets; fi is an indicator variable equal to one for firm observations reporting foreign income (pifo), and zero otherwise; and $R\&D$ is $R\&D$ expense (xrd) divided by lagged total assets.
DTAX	A firm's residual from the following regression based on Frank et al. (2009), estimated by industry and year: $PERMDIFF_{i,t} = \beta_0 + \beta_1 intang_{i,t} + \beta_2 equin_{i,t} + \beta_3 mi_{i,t} + \beta_4 txs_{i,t} + \beta_5 \Delta nol_{i,t} + \beta_6 PERMDIFF_{i,t-1} + \varepsilon_{i,t}$, where $PERMDIFF_{i,t}$ is calculated as $(pi - ((txfed + txfo)/str)) - (txdi/str)$ scaled by lagged assets (at). pi is pretax book income; str is the statutory tax rate in year t ; $txfed$ is the current federal tax expense; $txfo$ is the current foreign tax expense; $txdi$ is the deferred tax expense; $intang$ is goodwill and other intangible assets ($intan$) scaled by lagged assets (at); $equin$ is income (loss) reported under the equity method ($esub$) scaled by lagged assets (at); mi is income (loss) attributable to minority interest (mii) scaled by lagged assets (at); txs is current state tax expense scaled by lagged assets; Δnol is change in net operating loss carryforwards ($tlcf$) scaled by lagged assets (at).
PERMDIFF	ETR differential is $(pi - ((txfed + txfo)/str)) - (txdi/str)$ scaled by lagged assets (at), where pi is pretax book income, str is the statutory tax rate in year t , $txfed$ is the current federal tax expense, $txfo$ is the current foreign tax expense, and $txdi$ is the deferred tax expense. Based on Frank et al. (2009) and Kim et al. (2011).
<i>Law Changes</i>	
Good Faith	An indicator variable set to one if the state in which a firm is headquartered has adopted Good Faith Exception by year t and zero otherwise.
Implied Contract	An indicator variable set to one if the state in which a firm is headquartered has adopted the Implied Contract Exception by year t and zero otherwise.
Public Policy	An indicator variable set to one if the state in which a firm is headquartered has adopted the Public Policy exception by year t and zero otherwise.
Right-to-work Law	An indicator variable set to one if a firm is headquartered in a state that has passed Right-to-Work laws by year t and zero otherwise.
<i>Firm characteristics</i>	

Return on Assets	Pre-tax income (pi) in year t scaled by lagged asset (at).
Leverage	Long-term debt (dltt) in year t scaled by lagged asset (at).
Loss Carry Forward	A dummy variable equals to one if loss carry forward (tlcf) in year t is larger than zero, and zero otherwise.
Change in Loss Carry Forward	The change in loss carry forward (tlcf) from year t-1 to year t scaled by lagged assets (at).
Foreign Income	Foreign income (pifo) in year t scaled by lagged assets (at).
Net Property Plant and Equipment	Net property, plant, and equipment (ppent) scaled by lagged assets (at).
Intangible Assets	Intangible assets (intan) in year t scaled by lagged asset (at).
Equity Income in Earnings	Equity income in earnings (esub) in year t scaled by lagged asset (at).
Firm Size	The natural logarithm of total assets (at) at the beginning of year t.
Tobin's Q	Total assets (at) plus market value of equity (prcc*csho) minus book value of equity (ceq) minus deferred taxes (txdb) scaled by lagged assets (at).
Cash Flow	Income before extraordinary items (ib) plus depreciation and amortization (dp) scaled by lagged assets (at).
Cash Flow Volatility	The standard deviation of the ratio of income before extraordinary items plus depreciation and amortization to book assets over the past four years.
Dividend	An indicator variable equal to one if a firm pays common dividends (dvc), and zero otherwise.
Short-term Debt	Short-term debt (dlc) in year t scaled by lagged asset (at).
WW Index	Equals $-0.091 [(ib + dp)/at] - 0.062[\text{indicator set to one if } dvc + dvp \text{ is positive, and zero otherwise}] + 0.021[dltt/at] - 0.044[\log(at)] + 0.102[\text{average industry sales growth, estimated separately for each three-digit SIC industry and each year, with sales growth defined as above}] - 0.035[\text{sales growth}]$. See Whited and Wu (2006).
SA Index	Equals $-0.737\text{Size} + 0.043\text{Size}^2 - 0.040\text{Age}$; Size equals the log of inflation-adjusted Compustat item at; Age is the number of years the firm is listed with a non-missing stock price on Compustat. See Hadlock and Pierce (2010).
Z-Score	Modified Altman's (1968) Z-score, calculated as : $(1.2(wcap/at) + 1.4(re/at) + 3.3(ebit/at) + (sale/at))$.
R&D	R&D expenditures (xrd) scaled by lagged total assets (at).
Expected default Frequency (EDF)	A simplified version of Merton (1974) structural distance-to-default model. See Bharath and Shumway (2008).
Union Coverage	Union coverage density, measured by the number of unionized firms to total firms in each state. Source: https://www.unionstats.com/ .
Union Membership	Union membership density, measured by the ratio of unionized workers to total employees in each state. Source: https://www.unionstats.com/ .
Political Balance	The fraction of a state's Congress members in the U.S. House of Representatives that belong to the Democratic Party in a given year.
Log(State Per Capita GDP)	State Per Capita GDP is a state's GDP (in thousands) divided by its total population.

Table 1
State-Level Wrongful Discharge Laws

This table reports the month and year when each state passed good faith, implied contract, and public policy exceptions to the traditional employment at-will rule.

State	Month/Year Good Faith Exception Passed	Month/Year Implied Contract Exception Passed	Month/Year Public Policy Exception Passed
Alabama		7/1987	
Alaska	5/1983	5/1983	2/1986
Arizona	6/1985	6/1983(Reversed 4/1984)	6/1985
Arkansas		6/1984	3/1980
California	10/1980	3/1972	9/1959
Colorado		10/1983	9/1985
Connecticut	6/1980	10/1985	1/1980
Delaware	4/1992		3/1992
Florida			
Georgia			
Hawaii		8/1986	10/1982
Idaho	8/1989	4/1977	4/1977
Illinois		12/1974	12/1978
Indiana		8/1987	5/1973
Iowa		11/1987	7/1985
Kansas		8/1984	6/1981
Kentucky		8/1983	11/1983
Louisiana	1/1998		
Maine		11/1977	
Maryland		1/1985	7/1981
Massachusetts	7/1977	5/1988	5/1980
Michigan		6/1980	6/1976
Minnesota		4/1983	11/1986
Mississippi		6/1992	7/1987
Missouri		1/1983 (Reversed 2/1988)	11/1985
Montana	1/1982	6/1987	1/1980
Nebraska		11/1983	11/1987
Nevada	2/1987	8/1983	1/1984
New Hampshire	2/1974 (Reversed 5/1980)	8/1988	2/1974
New Jersey		5/1985	7/1980
New Mexico		2/1980	7/1983
New York		11/1982	
North Carolina			5/1985
North Dakota		2/1984	11/1987
Ohio		4/1982	3/1990
Oklahoma	5/1985 (Reversed 2/1989)	12/1976	2/1989
Oregon		3/1978	6/1975
Pennsylvania			3/1974
Rhode Island			
South Carolina		6/1987	11/1985
South Dakota		4/1983	12/1988
Tennessee		11/1981	8/1984
Texas		4/1985	6/1984
Utah	3/1989	5/1986	3/1989
Vermont		8/1985	9/1986
Virginia		9/1983	6/1985
Washington		8/1977	7/1984
West Virginia		4/1986	7/1978
Wisconsin		6/1985	1/1980
Wyoming	1/1994	8/1985	7/1989

Table 2
Summary Statistics for Main Variables

This table reports summary statistics for the main variables in the regression models. Panel A presents the summary statistics for the full sample. Panel B reports univariate results comparing the mean values of variables for the treatment (firms headquartered in states that eventually adopt the Good Faith Exception) and control firms (firms headquartered in states that never adopt this law). In Panel B, *, **, and *** in the column labeled Differences indicate significance at the 10%, 5%, and 1% levels, respectively, for a t-test of whether the two samples have equal means. The sample consists of Compustat non-financial firms over the 1970 to 2000 period and includes 57,467 firm-year observations. Continuous variables, except economic variables, are winsorized at their 1st and 99th percentiles. Appendix A provides variable definitions.

<i>Panel A: Summary Statistics for Full Sample</i>						
	Obs.	Mean	Std. Dev.	P25	Median	P75
Main Dependent Variable:						
DDBTD	57,467	-0.006	0.174	-0.036	0.021	0.077
Main Explanatory Variable:						
Good Faith	57,467	0.287	0.453	0.000	0.000	1.000
Control Variables:						
Implied Contracts	57,467	0.745	0.436	0.000	1.000	1.000
Public Policy	57,467	0.742	0.437	0.000	1.000	1.000
Return on Assets	57,467	0.031	0.242	-0.009	0.069	0.147
Leverage	57,467	0.214	0.248	0.013	0.139	0.322
Loss Carry Forward	57,467	0.267	0.443	0.000	0.000	1.000
Change in Loss						
Carry Forward	57,467	0.037	0.195	0.000	0.000	0.000
Foreign Income	57,467	0.004	0.018	0.000	0.000	0.000
Net Property Plant						
and Equipment	57,467	0.348	0.290	0.119	0.271	0.508
Intangible Assets	57,467	0.086	0.171	0.000	0.005	0.090
Equity Income in						
Earnings	57,467	0.000	0.005	0.000	0.000	0.000
Firm Size	57,467	4.719	2.097	3.178	4.544	6.048
Tobin's Q	57,467	2.178	3.137	1.037	1.385	2.213

Table 2 - continued
Summary Statistics for Main Variables

<i>Panel B: Comparing Sample Means for Treatment and Control Firms</i>			
	Treatment Sample (Obs.= 16,514)	Control Sample (Obs.= 40,953)	Differences
	(1)	(2)	(3)
DDBTD	-0.032	0.004	-0.036***
Implied Contracts	0.727	0.752	-0.025***
Public Policy	0.834	0.705	0.129***
Return on Assets	-0.002	0.044	-0.045***
Leverage	0.188	0.224	-0.036***
Loss Carry Forward	0.317	0.248	0.069***
Change in Loss Carry Forward	0.057	0.029	0.028***
Foreign Income	0.005	0.004	0.000
Net Property Plant and Equipment	0.301	0.366	-0.065***
Intangible Assets	0.086	0.086	0.000
Equity Income in Earnings	0.000	0.001	-0.000***
Firm Size	4.350	4.868	-0.518***
Tobin's Q	2.599	2.008	0.590***

Table 3
Determinants of the Adoption of Good Faith Exceptions

This table reports the results from a Cox proportional hazard model where the “failure event” is the adoption of the Good Faith Exception in a state, i.e. once a state adopts the law, it drops from the sample. The sample period is from 1970 to 2000. All explanatory variables are lagged by one year. Robust standard errors clustered at the state level are reported in parentheses. Coefficients marked with *, **, and *** are significant at 10%, 5%, and 1%, respectively. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Dependent Variable: Good Faith = 1		
	(1)	(2)	(3)
State-DDBTD	3.056 (10.173)	3.231 (7.585)	1.560 (7.114)
Implied Contract		-1.220 (0.809)	-1.517 (0.939)
Public Policy		0.571 (0.878)	0.270 (0.913)
GDP Growth		9.193 (8.367)	6.158 (8.740)
Real GDP		-0.331 (1.122)	0.200 (1.730)
Unemployment Rate		-0.033 (0.107)	0.141 (0.155)
Population		-0.144 (0.957)	-0.647 (1.520)
Union Coverage		-0.010 (0.034)	-0.034 (0.036)
Political Balance			-0.427** (0.189)
Observations	875	875	834
Adj R2	0.003	0.072	0.128

Table 4
The Good Faith Exception and Tax Aggressiveness

This table reports results from OLS regressions of tax aggressiveness on the indicator for the recognition of the Good Faith Exception. The sample period is from 1970 to 2000. The dependent variable is *DDBTD*, the Desai and Dharmapala (2006)'s residual book-tax difference. *Good Faith* is an indicator variable set to one if the state in which a firm is headquartered has adopted the Good Faith Exception by year *t* and zero otherwise. All variable definitions are defined formally in Appendix A. Continuous variables are winsorized at their 1st and 99th percentiles. Robust standard errors clustered at the state level are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Dependent Variable: DDBTD		
	(1)	(2)	(3)
Good Faith	-0.012*** (0.004)	-0.020*** (0.004)	-0.017*** (0.004)
Implied Contract		-0.002 (0.003)	-0.001 (0.003)
Public Policy		0.006* (0.003)	0.005 (0.003)
Return on Asset		0.340*** (0.009)	0.344*** (0.010)
Leverage		-0.010** (0.004)	-0.011*** (0.004)
Loss Carry Forward		0.026*** (0.002)	0.026*** (0.002)
Change in Loss Carry Forward		-0.041*** (0.007)	-0.041*** (0.006)
Foreign Income		-0.087 (0.061)	-0.083 (0.063)
Net Property Plant and Equipment		0.009 (0.008)	0.007 (0.008)
Intangible Assets		-0.037*** (0.008)	-0.036*** (0.008)
Equity Income in Earnings		-0.430*** (0.125)	-0.392*** (0.131)
Firm Size		0.026*** (0.002)	0.026*** (0.002)
Tobin's Q		-0.004*** (0.001)	-0.004*** (0.001)
Year Fixed Effects	Yes	Yes	No
Firm Fixed Effects	Yes	Yes	Yes
Industry*Year Fixed Effects	No	No	Yes
Observations	57,467	57,467	57,467
Adj R2	0.557	0.649	0.653

Table 5
The Timing of Changes in Firms' Tax Aggressiveness Around Adoptions of the Good Faith Exception

This table reports results from OLS regressions of tax aggressiveness on the indicator for the recognition of the Good Faith Exception. The sample period is from 1970 to 2000. The dependent variable is *DDBTD*, the Desai and Dharmapala (2006)'s residual book-tax difference. *Good Faith Adoption*⁻¹, *Good Faith Adoption*⁰, *Good Faith Adoption*⁺¹ and *Good Faith Adoption*⁺² are equal to one if the firm is headquartered in a state that will adopt the Good Faith Exception in one year, adopts the Good Faith Exception in the current year, adopted the Good Faith Exception one year ago, adopted the Good Faith Exception two years ago, respectively, and zero otherwise. Control variables include those in Table 4. All variables are as defined in Appendix A. Continuous variables are winsorized at their 1st and 99th percentiles. Robust standard errors clustered at the state level are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Dependent Variable: DDBTD		
	(1)	(2)	(3)
Good Faith Adoption ⁻¹	0.001 (0.008)	-0.005 (0.006)	-0.001 (0.006)
Good Faith Adoption ⁰	0.001 (0.006)	-0.010* (0.006)	-0.008 (0.005)
Good Faith Adoption ⁺¹	-0.006 (0.009)	-0.015** (0.006)	-0.012** (0.005)
Good Faith Adoption ⁺²	-0.017*** (0.006)	-0.026*** (0.005)	-0.021*** (0.006)
Control Variables	No	Yes	Yes
Year Fixed Effects	Yes	Yes	No
Firm Fixed Effects	Yes	Yes	Yes
Industry*Year Fixed Effects	No	No	Yes
Observations	57,467	57,467	57,467
Adj R ²	0.557	0.649	0.653

Table 6
Propensity-Score Matching

This table reports results from OLS regressions of tax aggressiveness on the indicator for the recognition of the Good Faith Exception as well as the timing of state courts' adoptions of the Good Faith based on a propensity-score matched sample. The sample period is from 1970 to 2000. The dependent variable is *DDBTD*, the Desai and Dharmapala (2006)'s residual book-tax difference. *Good Faith Adoption*⁻¹, *Good Faith Adoption*⁰, *Good Faith Adoption*⁺¹ and *Good Faith Adoption*⁺² are equal to one if the firm is headquartered in a state that will adopt the *Good Faith* in one years, adopts the *Good Faith* in the current year, adopted the *Good Faith* one year ago, adopted the *Good Faith* two years ago, respectively, and zero otherwise. Control variables include those in Table 4. All variables are as defined in Appendix A. Continuous variables are winsorized at their 1st and 99th percentiles. Robust standard errors clustered at the state level are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Dependent Variable: DDBTD	
	(1)	(2)
Good Faith	-0.020** (0.009)	
Good Faith Adoption ⁻¹		-0.003 (0.008)
Good Faith Adoption ⁰		-0.017** (0.006)
Good Faith Adoption ⁺¹		-0.020* (0.010)
Good Faith Adoption ⁺²		-0.024* (0.013)
Control Variables	Yes	Yes
Firm Fixed Effects	Yes	Yes
Industry*Year Fixed Effects	Yes	Yes
Observations	3,687	3,687
Adj R ²	0.367	0.367

Table 7
The Effect on Distress Risk

This table reports results from OLS regressions of distress risk on the indicator for the recognition of the Good Faith Exception as well as the timing of state courts' adoptions of the Good Faith. The sample period is from 1970 to 2000. *Zscore* is modified Altman's (1968) Z-score. *Default Risk* is computed using Moody's KMV model. *Good Faith Adoption*⁻¹, *Good Faith Adoption*⁰, *Good Faith Adoption*⁺¹ and *Good Faith Adoption*⁺² are equal to one if the firm is headquartered in a state that will adopt the *Good Faith* in one years, adopts the *Good Faith* in the current year, adopted the *Good Faith* one year ago, adopted the *Good Faith* two years ago, respectively, and zero otherwise. Control variables include those in Table 4. All variables are as defined in Appendix A. Continuous variables are winsorized at their 1st and 99th percentiles. Robust standard errors clustered at the state level are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Dependent Variable:	Zscore	Default Risk	Zscore	Default Risk
	(1)	(2)	(3)	(4)
Good Faith	-0.124** (0.047)	0.015*** (0.005)		
Good Faith Adoption ⁻¹			-0.031 (0.037)	-0.001 (0.007)
Good Faith Adoption ⁰			0.007 (0.045)	-0.006 (0.009)
Good Faith Adoption ⁺¹			0.012 (0.049)	-0.002 (0.011)
Good Faith Adoption ⁺²			-0.180*** (0.057)	0.024*** (0.008)
Control Variables	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes
Industry*Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	78,933	71,752	78,933	71,752
Adj R ²	0.790	0.460	0.790	0.460

Table 8
Cross-sectional Analysis: Financial Distress Risk

This table reports results from OLS regressions of DDBTD on indicators for the timing of state courts' adoptions of the Good Faith. The sample period is from 1970 to 2000. The dependent variable is *DDBTD*, the Desai and Dharmapala (2006)'s residual book-tax difference. *Good Faith* is an indicator variable set to one if the state in which a firm is headquartered has adopted the Good Faith Exception by year *t* and zero otherwise. *Zscore* is modified Altman's (1968) Z-score. *Estimated Default Frequency* is Bharath and Shumway (2008)'s EDF measure, a simplified version of Merton (1974) structural distance-to-default model. *Short-term Debt* is measured as short-term debt (*dlc*) scaled by lagged asset (*at*). *Cash Flow Volatility* is the standard deviation of the ratio of income before extraordinary items plus depreciation and amortization to book assets over the past four years. Standard errors are corrected for heteroskedasticity and clustering at the state level (standard errors are in parentheses). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Dependent Variable: DDBTD			
	(1)	(2)	(3)	(4)
Good Faith × Z-Score	0.010*** (0.003)			
Good Faith × Estimated Default Frequency		-0.103** (0.039)		
Good Faith × Short-Term Debt			-0.072** (0.033)	
Good Faith × Cash Flow Volatility				-0.489** (0.203)
Cohort*Year Fixed Effects	Yes	Yes	Yes	Yes
Cohort*Firm Fixed Effects	Yes	Yes	Yes	Yes
Industry*Year Fixed Effects	Yes	Yes	Yes	Yes
State*Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	226,532	225,333	253,069	254,177
Adj R ²	0.453	0.443	0.456	0.455

Table 9
Cross-sectional Analysis: Financial Constraints

This table reports results from OLS regressions of DDBTD on indicators for the timing of state courts' adoptions of the Good Faith. The sample period is from 1970 to 2000. The dependent variable is *DDBTD*, the Desai and Dharmapala (2006)'s residual book-tax difference. *Good Faith* is an indicator variable set to one if the state in which a firm is headquartered has adopted the Good Faith Exception by year *t* and zero otherwise. *No Dividend* is an indicator variable equal to one if a firm does not pay common dividends (*dvc*), and zero otherwise. *WW Index* is constructed following Whited and Wu (2006). *SA Index* is constructed following Hadlock and Pierce (2010). Standard errors are corrected for heteroskedasticity and clustering at the state level (standard errors are in parentheses). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Dependent Variable: DDBTD		
	(1)	(2)	(3)
Good Faith × No Dividend	-0.032*** (0.009)		
Good Faith × WW Index		-0.001* (0.001)	
Good Faith × SA Index			-0.024*** (0.007)
Cohort*Year Fixed Effects	Yes	Yes	Yes
Cohort*Firm Fixed Effects	Yes	Yes	Yes
Industry*Year Fixed Effects	Yes	Yes	Yes
State*Year Fixed Effects	Yes	Yes	Yes
Observations	254,395	237,931	253,236
Adj R ²	0.455	0.433	0.455

Table 10
Alternative Explanations

This table reports results from OLS regressions of effects of employment protection on tax aggressiveness after controlling for alternative explanations. Panel A presents the results of effects of employment protection on tax aggressiveness after controlling for labor union coverage. The dependent variable is *DDBTD*, the Desai and Dharmapala (2006)'s residual book-tax difference. *Good Faith* is an indicator variable set to one if the state in which a firm is headquartered has adopted the Good Faith Exception by year *t* and zero otherwise. *Right-to-Work* is an indicator variable set to one if a firm is headquartered in a state that has passed right-to-work laws by year *t* and zero otherwise. *Union coverage* is measured by the number of unionized firms to total firms in each state. *Union membership* is measured by the ratio of unionized workers to total employees in each state. In Columns (2)-(3), the sample spans the 1983-2000 since *union data* span the 1983-2000. Panel B reports the results of effects of employment protection on tax aggressiveness after controlling for alternative explanations. Column (1)-(2) examine the effect of Good Faith Exception for firms that experience a decrease or an increase in capital expenditures (*capx*) from year *t-1* to year *t*. Column (3)-(4) examine the effect of Good Faith Exception for firms that experience a negative or a positive correlation between the changes in tax aggressiveness and those in capital expenditures. Column (5)-(6) examine the effect of Good Faith Exception for firms with or without patent applications. Standard errors are corrected for heteroskedasticity and clustering at the state level (standard errors are in parentheses). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Labor Union

Dependent Variable: DDBTD			
	(1)	(2)	(3)
Good Faith	-0.013*** (0.004)	-0.007 (0.012)	-0.007 (0.010)
Good Faith × Right-to-work Law	0.003 (0.006)		
Good Faith × Union Coverage		-0.000 (0.001)	
Good Faith × Union Membership			-0.000 (0.001)
Cohort*Year Fixed Effects	Yes	Yes	Yes
Cohort*Firm Fixed Effects	Yes	Yes	Yes
Industry*Year Fixed Effects	Yes	Yes	Yes
Observations	254,398	244,150	244,150
Adj R ²	0.443	0.443	0.443

Panel B: The Impact of Firm Investment

Dependent Variable: DDBTD						
	Decrease	Increase	Negative correlation	Positive correlation	Zero patents	Non-zero patents
	(1)	(2)	(3)	(4)	(5)	(6)
Good Faith	-0.016** (0.008)	-0.018*** (0.004)	-0.021** (0.008)	-0.022*** (0.006)	-0.012** (0.005)	-0.040 (0.044)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry*Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	19,922	31,130	21,137	23,022	43,474	10,822
Adj R ²	0.667	0.640	0.673	0.670	0.614	0.770

Table 11
Alternative Measures of Corporate Tax Aggressiveness

This table reports results from OLS regressions of alternative measures of tax aggressiveness on the indicator for the recognition of the Good Faith Exception and on indicators for the timing of state courts' adoptions of the Good Faith. The sample period is from 1970 to 2000. *BTD* is total book-tax difference, the book income less taxable income scaled by lagged assets (at). *SHELTER* is estimated probability that a firm engages in a tax shelter based on Wilson's (2009) tax-sheltering model. *DTAX* is a firm's residual from the regression based on Frank et al. (2009). *PERMDIFF* is the ETR differential based on Frank et al. (2009) and Kim et al. (2011). *Good Faith* is an indicator variable set to one if the state in which a firm is headquartered has adopted the Good Faith Exception by year *t* and zero otherwise. *Good Faith Adoption*⁻¹, *Good Faith Adoption*⁰, *Good Faith Adoption*⁺¹ and *Good Faith Adoption*⁺² are equal to one if the firm is headquartered in a state that will adopt the *Good Faith* in one years, adopts the *Good Faith* in the current year, adopted the *Good Faith* one year ago, adopted the *Good Faith* two years ago, respectively, and zero otherwise. Control variables include those in Table 4. All variables are as defined in Appendix A. Continuous variables are winsorized at their 1st and 99th percentiles. Standard errors are corrected for heteroskedasticity and clustering at the state level (standard errors are in parentheses). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Dependent Variable:	BTD (1)	SHELTER (2)	DTAX (3)	PERMDIFF (4)	BTD (5)	SHELTER (6)	DTAX (7)	PERMDIFF (8)
Good Faith	-0.006** (0.002)	-0.079*** (0.028)	-0.005** (0.002)	-0.005** (0.002)				
Good Faith Adoption ⁻¹					0.001 (0.005)	-0.018 (0.051)	-0.000 (0.004)	0.002 (0.003)
Good Faith Adoption ⁰					0.001 (0.004)	0.018 (0.044)	-0.006* (0.003)	-0.003 (0.004)
Good Faith Adoption ⁺¹					0.001 (0.004)	0.013 (0.048)	0.000 (0.004)	0.001 (0.003)
Good Faith Adoption ⁺²					-0.008** (0.003)	-0.116*** (0.034)	-0.006** (0.002)	-0.006** (0.003)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry*Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	69,395	69,273	45,818	45,818	69,395	69,273	45,818	45,818
Adj R ²	0.850	0.876	0.285	0.822	0.850	0.876	0.285	0.822

Table 12
Robustness Checks

This table reports robustness checks results of the effects of employment protection on tax aggressiveness. The sample period is from 1970 to 2000. *DDBTD* is the Desai and Dharmapala (2006)'s residual book-tax difference. *BTD* is total book-tax difference, the book income less taxable income scaled by lagged assets (at). *SHELTER* is estimated probability that a firm engages in a tax shelter based on Wilson's (2009) tax-sheltering model. *DTAX* is a firm's residual from the regression based on Frank et al. (2009). *PERMDIFF* is the ETR differential based on Frank et al. (2009) and Kim et al. (2011). *Good Faith* is an indicator variable set to one if the state in which a firm is headquartered has adopted the Good Faith Exception by year *t* and zero otherwise. Control variables include those in Table 4. All variables are as defined in Appendix A. Continuous variables are winsorized at their 1st and 99th percentiles. Standard errors are corrected for heteroskedasticity and clustering at the state level (standard errors are in parentheses). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

<i>Panel A: Cohort-Matching Sample</i>					
Dependent Variable:	DDBTD (1)	BTD (2)	SHELTER (3)	DTAX (4)	PERMDIFF (5)
Good Faith	-0.016*** (0.004)	-0.023*** (0.005)	-0.451*** (0.101)	-0.003*** (0.001)	-0.015*** (0.005)
Cohort*Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Cohort*Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes
Industry*Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	348,162	412,620	411,378	235,227	235,227
Adj R ²	0.439	0.455	0.526	0.118	0.489
<i>Panel B: Further controlling for Region*Year Fixed Effects</i>					
Dependent Variable:	DDBTD (1)	BTD (2)	SHELTER (3)	DTAX (4)	PERMDIFF (5)
Good Faith	-0.019*** (0.004)	-0.007*** (0.002)	-0.089*** (0.025)	-0.004* (0.002)	-0.007*** (0.002)
Control Variables	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes
Industry*Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Region*Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	57,467	69,395	69,273	45,818	45,818
Adj R ²	0.653	0.850	0.876	0.286	0.822
<i>Panel C: Excluding CA and MA Headquartered Firms</i>					
Dependent Variable:	DDBTD (1)	BTD (2)	SHELTER (3)	DTAX (4)	PERMDIFF (5)
Good Faith	-0.019*** (0.005)	-0.008** (0.004)	-0.091** (0.041)	-0.004 (0.003)	-0.006* (0.003)
Control Variables	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes
Industry*Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	45,517	55,962	55,872	36,725	36,725
Adj R ²	0.647	0.848	0.878	0.281	0.810

Table 12 - continued
Robustness Checks

<i>Panel D: Sales growth $\leq 100\%$</i>					
Dependent Variable:	DDBTD (1)	BTD (2)	SHELTER (3)	DTAX (4)	PERMDIFF (5)
Good Faith	-0.014*** (0.004)	-0.006** (0.003)	-0.088*** (0.030)	-0.003* (0.002)	-0.005** (0.002)
Control Variables	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes
Industry*Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	52,827	64,682	64,568	43,307	43,307
Adj R ²	0.622	0.824	0.868	0.285	0.802
<i>Panel E: Excluding firms that changed headquarters during sample period</i>					
Dependent Variable:	DDBTD (1)	BTD (2)	SHELTER (3)	DTAX (4)	PERMDIFF (5)
Good Faith	-0.021*** (0.007)	-0.007** (0.003)	-0.091** (0.043)	-0.006*** (0.002)	-0.006** (0.002)
Control Variables	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes
Industry*Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	43,867	53,514	53,421	34,956	34,956
Adj R ²	0.651	0.839	0.874	0.278	0.812

Employment Protection and Tax Aggressiveness: Evidence from Wrongful Discharge Laws

Internet Appendix

Table A1
The Good Faith Exception and Tax Aggressiveness –
Controlling for Lagged DDBTD

This table reports results from OLS regressions of tax aggressiveness on the indicator for the recognition of the Good Faith Exception when controlling for lagged *DDBTD*. The sample period is from 1970 to 2000. The dependent variable is *DDBTD*, the Desai and Dharmapala (2006)'s residual book-tax difference. *Good Faith* is an indicator variable set to one if the state in which a firm is headquartered has adopted the Good Faith Exception by year t and zero otherwise. Control variables include those in Table 4. All variables are as defined in Appendix A. Continuous variables are winsorized at their 1st and 99th percentiles. Robust standard errors clustered at the state level are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Dependent Variable: DDBTD		
	(1)	(2)	(3)
Good Faith	-0.018*** (0.004)	-0.024*** (0.005)	-0.019*** (0.005)
Lagged DDBTD	0.167*** (0.013)	0.099*** (0.010)	0.090*** (0.009)
Controls	No	Yes	Yes
Year Fixed Effects	Yes	Yes	No
Firm Fixed Effects	Yes	Yes	Yes
Industry*Year Fixed Effects	No	No	Yes
Observations	45,031	45,031	45,031
Adj R2	0.587	0.670	0.672

Table A2
Statistics of Propensity Score Matching

This table presents statistics of propensity score matching. Panel A presents parameter estimates from the logistic model used to estimate propensity scores for firms in the treatment and control groups. Standard errors are in parentheses. Panel B reports the distribution of estimated propensity scores for the treatment firms, control firms, and the difference in estimated propensity scores post matching. Panel C presents statistics of the post-match differences in propensity score matching attached to the Differences-in-Differences analysis. Column (1) presents the sample average of firm characteristics in the treated group. Column (2) presents the sample average of firm characteristics in the control group. Column (3) presents the difference between Columns (1) and (2). Column (4) presents the t-test value of the difference test between Columns (1) and (2). Definitions of all these variables are provided in Appendix A.

<i>Panel A: Pre-match regression and post-match diagnostic regression</i>		
	Pre-match (1)	Post-match (2)
Implied Contract	-0.069 (0.107)	0.459 (0.325)
Public Policy	0.209** (0.103)	-0.569* (0.299)
Return on Asset	0.387 (0.284)	-0.282 (0.791)
Leverage	0.014 (0.227)	-0.149 (0.551)
Loss Carry Forward	0.267** (0.111)	0.290 (0.287)
Change in Loss Carry Forward	-0.223 (0.361)	-0.178 (1.024)
Foreign Income	-0.055 (4.888)	7.233 (9.091)
Net Property Plant and Equipment	0.027 (0.212)	-0.309 (0.503)
Intangible Assets	0.188 (0.486)	-0.388 (0.808)
Equity Income in Earnings	-13.185 (8.332)	-9.774 (23.403)
Firm Size	-0.101*** (0.026)	-0.006 (0.083)
Tobin's Q	0.004 (0.017)	0.009 (0.072)
Industry Fixed Effects	Yes	Yes
Cohort Fixed Effects	Yes	Yes
Observations	30,865	396
p-value of χ^2	0.000	0.846
Pseudo_R2	0.199	0.053

Table A2
Statistics of Propensity Score Matching - continued

<i>Panel B: Estimated Propensity Score Distributions</i>								
	Obs. (1)	Mean (2)	Std. Dev. (3)	Min (4)	P25 (5)	Median (6)	P75 (7)	Max (8)
Treated	198	0.053	0.053	0.001	0.011	0.036	0.076	0.326
Control	198	0.053	0.053	0.001	0.011	0.036	0.076	0.321
Difference		0.000	0.001	0.000	0.000	0.000	0.000	0.005
<i>Panel C: Post-Match Differences</i>								
	Treatment (N=198)		Control (N=198)		Differences		T-value	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Implied Contracts		0.338	0.281		0.057		1.02	
Public Policy		0.449	0.476		-0.027		-0.44	
Return on Assets		0.062	0.065		-0.002		-0.10	
Leverage		0.245	0.252		-0.008		-0.23	
Loss Carry Forward		0.313	0.267		0.046		0.84	
Change in Loss Carry Forward		0.022	0.023		-0.001		-0.04	
Foreign Income		0.002	0.003		0.000		-0.07	
Net Property Plant and Equipment		0.424	0.452		-0.028		-0.74	
Intangible Assets		0.053	0.059		-0.006		-0.36	
Equity Income in Earnings		0.001	0.001		0.000		-0.44	
Firm Size		4.203	4.283		-0.080		-0.34	
Tobin's Q		2.153	1.938		0.215		0.61	

Table A3

Cross-sectional Analysis with Control Variables: Financial Distress Risk

This table reports results from OLS regressions of DDBTD on indicators for the timing of state courts' adoptions of the Good Faith. The sample period is from 1970 to 2000. The dependent variable is *DDBTD*, the Desai and Dharmapala (2006)'s residual book-tax difference. *Good Faith* is an indicator variable set to one if the state in which a firm is headquartered has adopted the Good Faith Exception by year *t* and zero otherwise. *Zscore* is modified Altman's (1968) Z-score. *Estimated Default Frequency* is Bharath and Shumway (2008)'s EDF measure, a simplified version of Merton (1974) structural distance-to-default model. *Short-term Debt* is measured as short-term debt (dlc) scaled by lagged asset (at). *Cash Flow Volatility* is the standard deviation of the ratio of income before extraordinary items plus depreciation and amortization to book assets over the past four years. Standard errors are corrected for heteroskedasticity and clustering at the state level (standard errors are in parentheses). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Dependent Variable: DDBTD			
	(1)	(2)	(3)	(4)
Good Faith × Z-Score	0.022*** (0.004)			
Good Faith × Estimated Default Frequency		-0.111*** (0.034)		
Good Faith × Short-Term Debt			-0.165* (0.082)	
Good Faith × Cash Flow Volatility				-0.712*** (0.218)
Return on Asset	0.285*** (0.015)	0.283*** (0.015)	0.286*** (0.015)	0.286*** (0.015)
Leverage	-0.014*** (0.004)	-0.016*** (0.005)	-0.014** (0.005)	-0.014*** (0.005)
Loss Carry Forward	0.021*** (0.002)	0.019*** (0.002)	0.020*** (0.002)	0.020*** (0.002)
Change in Loss Carry Forward	-0.032*** (0.008)	-0.033*** (0.008)	-0.031*** (0.008)	-0.031*** (0.008)
Foreign Income	-0.096 (0.064)	-0.078 (0.066)	-0.105 (0.064)	-0.104 (0.063)
Net Property Plant and Equipment	0.035*** (0.008)	0.034*** (0.008)	0.032*** (0.008)	0.032*** (0.008)
Intangible Assets	-0.038*** (0.012)	-0.030** (0.013)	-0.035** (0.014)	-0.035** (0.014)
Equity Income in Earnings	-0.181 (0.174)	-0.184 (0.171)	-0.203 (0.165)	-0.207 (0.165)
Firm Size	0.021*** (0.001)	0.020*** (0.001)	0.020*** (0.001)	0.020*** (0.001)
Tobin's Q	-0.012*** (0.001)	-0.012*** (0.001)	-0.012*** (0.001)	-0.012*** (0.001)
Cohort*Year Fixed Effects	Yes	Yes	Yes	Yes
Cohort*Firm Fixed Effects	Yes	Yes	Yes	Yes
Industry*Year Fixed Effects	Yes	Yes	Yes	Yes
State*Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	178,565	176,948	199,185	199,891
Adj R ²	0.546	0.533	0.546	0.545

Table A4

Cross-sectional Analysis with Control Variables: Financial Constraints

This table reports results from OLS regressions of DDBTD on indicators for the timing of state courts' adoptions of the Good Faith. The sample period is from 1970 to 2000. The dependent variable is *DDBTD*, the Desai and Dharmapala (2006)'s residual book-tax difference. *Good Faith* is an indicator variable set to one if the state in which a firm is headquartered has adopted the Good Faith Exception by year *t* and zero otherwise. *Firm Size* is the natural logarithm of total assets (*at*) at the beginning of year *t*. *Dividend* is an indicator variable equal to one if a firm pays common dividends (*dvc*), and zero otherwise. *WW Index* is constructed following Whited and Wu (2006). *SA Index* is constructed following Hadlock and Pierce (2010). Standard errors are corrected for heteroskedasticity and clustering at the state level (standard errors are in parentheses). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Dependent Variable: DDBTD			
	(1)	(2)	(3)	(4)
Good Faith × Firm Size	0.017*** (0.005)			
Good Faith × Dividend		0.065*** (0.015)		
Good Faith × WW Index			-0.002*** (0.000)	
Good Faith × SA Index				-0.044*** (0.010)
Return on Asset	0.286*** (0.015)	0.286*** (0.015)	0.271*** (0.015)	0.286*** (0.015)
Leverage	-0.014** (0.005)	-0.014** (0.005)	-0.015*** (0.005)	-0.014** (0.005)
Loss Carry Forward	0.020*** (0.002)	0.020*** (0.002)	0.019*** (0.002)	0.020*** (0.002)
Change in Loss Carry Forward	-0.031*** (0.008)	-0.031*** (0.008)	-0.033*** (0.009)	-0.031*** (0.008)
Foreign Income	-0.105 (0.064)	-0.104 (0.064)	-0.087 (0.062)	-0.101 (0.065)
Net Property Plant and Equipment	0.032*** (0.008)	0.032*** (0.008)	0.036*** (0.009)	0.032*** (0.008)
Intangible Assets	-0.035** (0.014)	-0.034** (0.014)	-0.031** (0.014)	-0.035** (0.014)
Equity Income in Earnings	-0.200 (0.164)	-0.202 (0.164)	-0.133 (0.158)	-0.201 (0.164)
Firm Size	0.020*** (0.001)	0.020*** (0.001)	0.019*** (0.001)	0.020*** (0.001)
Tobin's Q	-0.012*** (0.001)	-0.012*** (0.001)	-0.012*** (0.001)	-0.012*** (0.001)
Cohort*Year Fixed Effects	Yes	Yes	Yes	Yes
Cohort*Firm Fixed Effects	Yes	Yes	Yes	Yes
Industry*Year Fixed Effects	Yes	Yes	Yes	Yes
State*Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	199,628	200,088	187,221	199,582
Adj R ²	0.546	0.546	0.518	0.546